
MuteBench: Modality Unavailability Tolerance Evaluation for Incomplete Multimodal Fusion

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Abstract

Multimodal physiological data powers clinical AI systems from intensive care units to wearable devices, but sensors routinely fail in practice. Two failure modes are common: modality missing, where an entire channel is absent, and within-modality missing, where a contiguous time segment is lost. No existing benchmark evaluates multiple fusion architectures under both failure modes at controlled severity levels across diverse clinical datasets. We present MuteBench, a benchmark covering 9 datasets from 7 clinical domains, 6 fusion architectures, and 2 missing-data modes over 125,000 samples. Through this benchmark, we find that architecture family is the strongest predictor of robustness, outweighing parameter count. Channel-independent models tolerate modality missing well but can be sensitive to within-modality missing, especially on short sequences. Curriculum modality dropout protects reliably only up to the maximum dropout rate used in training. We also find that channel count, sequence length, and modality alignment jointly determine which failure mode poses the greater threat. Finally, a PTB-XL case study suggests that diffusion-based imputation can improve downstream classification under within-modality missing, with the largest gains for models whose expert routing is most sensitive to corrupted inputs, though broader validation across datasets remains an open direction. MuteBench provides practitioners with concrete guidance for both selecting existing architectures and informing the design of future robust multimodal fusion methods.

1 Introduction

Multimodal physiological data is central to modern clinical AI [2, 37], combining ECG, EEG, PPG, and structured records to support disease prediction, severity assessment, and health monitoring [47, 33, 9]. Yet sensors routinely fail in practice. Leads detach, devices lose contact, and transmission errors drop segments of data [7, 41], resulting in two distinct failure modes. *Modality missing* occurs when an entire channel is absent [50], while *within-modality missing* refers to the loss of a contiguous time segment within an active channel [5]. Both failures arise during deployment and data collection alike, making incomplete multimodal data a common feature of clinical datasets.

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Table 1: Feature comparison of multimodal robustness benchmarks. MuteBench is the first to evaluate robustness under various missingness conditions at controlled severity levels across multiple fusion architectures and clinical physiological datasets.

Benchmark	Clinical	Physio. TS	Multi- dataset	Multi- arch	Mod. Missing	Within-mod. Missing	Severity Levels	Public Protocol
MultiBench [25]	✗	✗	✓	✓	✓	✗	✗	✓
BenchMD [46]	✓	✓	✓	✗	✗	✗	✗	✓
MC-BEC [6]	✓	✗	✓	✗	✗	✗	✗	✗
CLIMB [8]	✓	✓	✓	✗	✗	✗	✗	✓
M3Care [50]	✓	✓	✗	✗	✓	✗	✗	✓
MuteBench (Ours)	✓	✓	✓	✓	✓	✓	✓	✓

Despite this, most multimodal models are trained and evaluated assuming complete modality availability [45, 11], overlooking the incompleteness common in clinical practice. Even when data are complete, training dynamics tend to favor certain modalities over others [44, 16], so many multimodal models in practice rely on one dominant modality while leaving others underused. Missing modalities therefore worsen these existing imbalances rather than creating an entirely new challenge. Recent evidence further shows that the two failure modes produce inconsistent method rankings [1], yet no existing work jointly evaluates multiple architectures under both modes at controlled severity levels across diverse clinical signals (Table 1), leaving practitioners without clear guidance for building robust multimodal fusion systems.

To address this gap, we introduce the Modality Unavailability Tolerance Evaluation Benchmark (MuteBench), covering 9 datasets from 7 clinical domains, over 125,000 total samples across datasets ranging from 515 to 64,726 recordings, and 6 fusion models from three architecture families: channel-independent models, Mixture-of-Experts fusion models, and shared-specific decomposition models. All models are evaluated under two missingness conditions at controlled severity levels, yielding 270 evaluation configurations replicated across three independent random seeds (810 total runs). The datasets span channel counts from 2 to 78, sequence lengths from 48 to 3,000 time steps, and sampling rates from once-per-hour clinical observations to 4,000 Hz physiological waveforms, covering binary, multi-class, and multi-label tasks across all three modality alignment types. A framework-agnostic missingness library injects identical deterministic patterns across all codebases, ensuring that performance differences reflect architectural choices rather than data-pipeline artifacts. Our main contributions are:

- **The first benchmark to jointly evaluate robustness under various missingness conditions across multiple fusion architectures and clinical datasets.** MuteBench is the first to cross multiple axes simultaneously: 9 datasets from 7 clinical domains, 6 fusion architectures from three model families, and both modality missing and within-modality missing at controlled severity levels. A framework-agnostic missingness library and unified evaluation protocol ensure that observed performance differences reflect architecture, not pipeline artifacts.
- **Empirical insights on what drives robustness.** Benchmarking across all nine datasets reveals that architecture family, not parameter count, is the primary driver of robustness. Channel-independent models are most stable under modality missing, while shared-specific models best handle within-modality missing on homogeneous signals. Dataset channel count and sequence length jointly determine which failure mode is more damaging. Curriculum modality dropout provides strong protection, but only up to the missing rate covered during training.
- **Diffusion imputation as a potential remedy for within-modality missing.** In a PTB-XL case study with three models, diffusion-based imputation improves downstream classification under within-modality missing, with gains scaling with missing severity. The largest gains appear for models whose expert routing is most sensitive to corrupted inputs, while architectures that already handle within-modality gaps structurally benefit less. The benefit is negligible under modality missing, where an entirely absent channel provides little conditioning signal for reconstruction.

In light of these findings, our MuteBench provides a view with concrete, dataset-aware guidance for selecting fusion architectures under real-world missingness. Beyond evaluation, our findings directly inform the design of future robust multimodal fusion models by identifying structural properties

(modality alignment type, channel count, and sequence length) that govern sensitivity to each failure mode. Specifically, our results suggest that dataset properties such as channel count, sequence length, and modality alignment type should be treated as first-class design inputs when developing robust fusion architectures, rather than being handled post-hoc through generic missing-data strategies. Together, MuteBench and these findings establish the first systematic foundation for benchmarking and designing robust multimodal clinical fusion systems under realistic sensor failure conditions.

2 Related Work

Clinical Multimodal Benchmarks. Clinical multimodal benchmarks fall into two categories. The first covers medical question answering [18, 36], LLM and LVLM reasoning on clinical images and guidelines [40, 15, 24, 10, 17], and multimodal QA over EHR tables paired with radiology images [3]. These all focus on text-based or knowledge-driven reasoning without continuously sampled physiological signals. The second targets task-oriented clinical prediction by combining time series, waveforms, and structured features [6, 50, 46]. Liang et al. [25] introduce a general-purpose benchmark with robustness evaluation but do not separate modality missing from within-modality missing patterns and do not focus on clinical physiological signals. No existing benchmark systematically studies how different missing patterns interact with signal structure across multiple fusion architectures, leaving an important and unaddressed gap in multimodal clinical evaluation.

Missing Data in Clinical Time Series. Most multimodal healthcare prediction models are developed and benchmarked assuming complete modality availability [45], yet this assumption rarely holds in real clinical practice. Missing data in clinical physiological signals takes two forms: modality missing (an entire sensor stream absent) and within-modality missing (contiguous temporal gaps within an active channel). Single-modality methods address these challenges through missingness-aware recurrent models [5], continuous-time attention networks [34], and graph-based sensor-dropout encoders [51]. For multimodal settings, existing methods include generative reconstruction of absent modalities [26], shared-specific feature decomposition [43, 48], and prompt-based adaptation [22]. These methods are each proposed and evaluated in isolation on individual datasets, without a unified comparison across architectures or clinical signal structures.

Benchmark for Robustness. Two gaps remain. First, no benchmark jointly evaluates multiple fusion architectures under both missing patterns at controlled severity levels on clinical physiological signals. MultiBench [25] and MC-BEC [6] each address only one of these dimensions. Second, no benchmark accounts for how dataset structure (modality alignment type, channel count, and sequence length) interacts with missing pattern type to determine degradation. Without this, it is unclear which architectural choices suit which deployment scenario. We address both gaps with nine clinical datasets across three modality alignment structures, six fusion architectures, and two missing patterns at two severity levels. Table 1 details this comparison. As Section 4 shows, architecture rankings can reverse between the two failure modes, making isolated evaluation misleading.

3 Dataset

This section describes the datasets used in MuteBench. We select 9 datasets from 7 clinical domains (Figure 1, left), covering a broad range of real-world conditions, so that our robustness findings reflect general trends rather than dataset-specific artifacts. The following subsections describe our selection criteria and modality definitions.

3.1 Dataset Selection and Modality Definitions

We curate 9 datasets spanning 7 domains (e.g., ICU, cardiology, wearable activity tracking) and multiple task formats (binary, multi-class, multi-label) to measure both spatial and temporal robustness. This diversity ensures that the evaluated fusion strategies are tested across varied input structures and clinical objectives, enabling a thorough assessment of generalization across diverse clinical settings.

We categorize the datasets into three modality alignment types to isolate how structural alignment affects a model’s ability to recover missing information. **Type 1 (Homogeneous and Aligned)** datasets share the same format and time axis (e.g., 12-lead ECG), testing pure spatial compensation. **Type 2 (Heterogeneous and Aligned)** datasets have different physical domains but synchronized time axes (e.g., EEG and respiration), testing cross-domain temporal reasoning. **Type 3 (Heterogeneous**

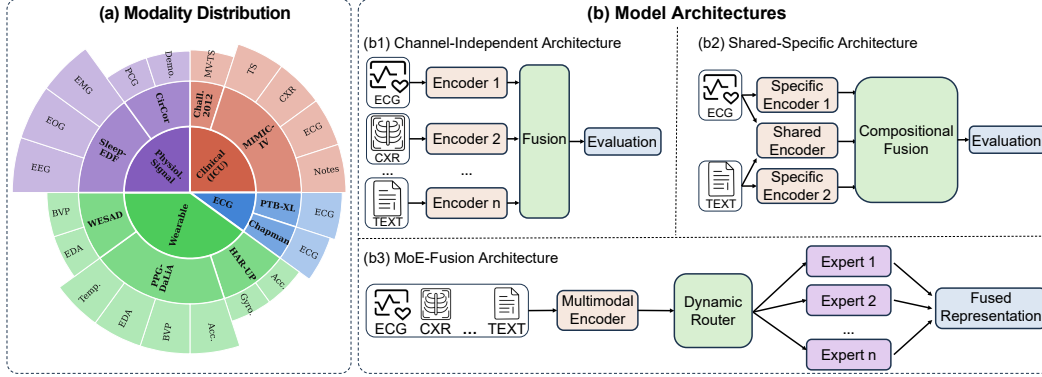


Figure 1: Overview of MuteBench. We evaluate 9 datasets spanning 7 clinical domains and three representative fusion architecture families: channel-independent models, Mixture-of-Experts fusion models, and shared-specific decomposition models, under two systematic missingness patterns at controlled severity levels.

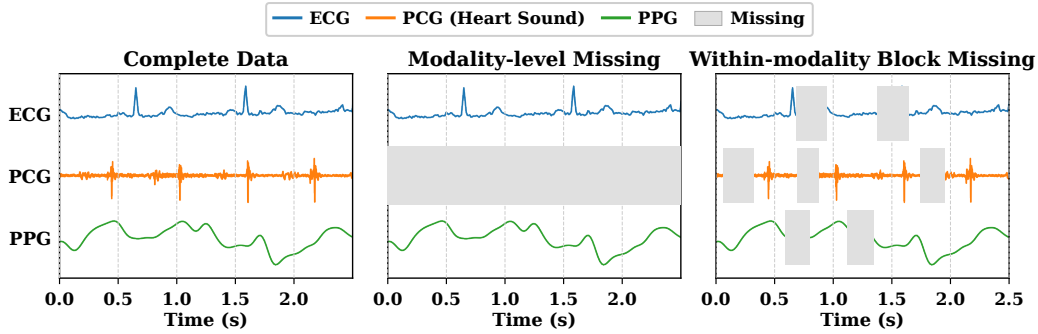


Figure 2: Evaluation of missing-data conditions. **Left (Complete):** Original, fully observed signals. **Middle (Modality missing):** Entire modalities (e.g., B and E) are dropped with probability p , simulating whole-sensor failures. **Right (Within-modality missing):** Contiguous time segments are masked independently per channel, simulating transient interruptions like motion artifacts.

and Unaligned) datasets lack both spatial and temporal synchronization (e.g., asynchronous clinical records combined with static metadata), testing models under the most challenging alignment conditions. Detailed selection criteria and formal modality definitions are in Appendix C.

3.2 Dataset Statistics

The benchmark spans 9 datasets from 7 clinical and physiological domains, totalling over 125,000 samples. Dataset sizes differ by more than two orders of magnitude: HAR-UP [27] is the smallest at 515 samples and PPG-DaLiA [30] the largest at 64,726. ICU datasets fall in the mid-range, with MIMIC-IV [20] contributing 5,100 patient episodes and Challenge-2012 [35] about 4,000 hospital stays. The cardiac datasets are similarly sized: PTB-XL [42] provides 21,837 recordings and Chapman-Shaoxing [52] 10,646. Task formats are equally diverse. Three datasets target binary classification: HAR-UP, MIMIC-IV, and Challenge-2012. Four require multi-class prediction with 3 to 9 categories: Sleep-EDF [21] distinguishes 5 sleep stages, PPG-DaLiA covers 9 activity types, WESAD [31] recognizes 3 affective states, and CirCor [28] grades 3 murmur severities. The remaining two, PTB-XL and Chapman-Shaoxing, use multi-label prediction with 5 and 7 diagnostic label groups per recording. All datasets are evaluated with Macro-AUROC as the primary metric and Macro-F1 reported alongside.

The datasets also vary substantially in input structure. Channel counts range from 2 in WESAD (wrist EDA and BVP sensors) to 78 in CirCor (64 mel-spectrogram bins plus 14 static demographic channels). Sequence lengths span from 48 time steps in MIMIC-IV and Challenge-2012 (hourly ICU

Table 2: Overview of the specific datasets evaluated in the benchmark. The table is sorted by modality type and task type.

Dataset	Domain	Samples	Channels	Steps	Modality	Task Type	Metric
HAR-UP	HAR	515	30	140	Type 1	Binary	AUROC
PTB-XL	Cardio.	21.8K	12	250	Type 1	Multi-label	AUROC
Chapman	Cardio.	10.6K	12	1000	Type 1	Multi-label	AUROC
Sleep-EDF	Neuro.	10.9K	5	3000	Type 2	Multi-class	AUROC
PPG-DaLiA	HAR	64.7K	9	256	Type 2	Multi-class	AUROC
WESAD	Affect	4.4K	2	480	Type 2	Multi-class	AUROC
MIMIC-IV	ICU	5.1K	30	48	Type 3	Binary	AUROC
Challenge-2012	ICU	4K	42	48	Type 3	Binary	AUROC
CirCor	Cardiac	3.1K	78	625	Type 3	Multi-class	AUROC

Table 3: Summary of baseline model architectures and fusion strategies.

Model	Backbone Encoder	Key Fusion Strategy	Sparse MoE
CLIMB	Transformer	Channel-independent	No
Flex-MoE	Transformer	G-/S-Router + Modality Bank	Yes
MIRA	Neural ODE	Frequency-domain MoE	Yes
ShaSpec	CNN/MLP	Shared-specific Decomposition	No
FuseMoE	Transformer	Sparse MoE + mTAND	Yes
Maestro	Sparse Attention (per-modal)	Sparse Attn + Curriculum Dropout	Yes

bins) to 3,000 steps in Sleep-EDF (30-second windows at 100 Hz). Other examples include PTB-XL and Chapman-Shaoxing with 12 ECG leads at $T=250$ and $T=1,000$, PPG-DaLiA with 9 wearable channels at $T=256$, and HAR-UP with 30 IMU channels from 5 body-worn sensors at $T=140$. Sampling rates vary from once-per-hour clinical observations to 4,000 Hz phonocardiogram waveforms. Challenge-2012 has extreme temporal sparsity, with only $\approx 13.9\%$ of per-cell measurements observed, making it the most irregularly sampled dataset in the benchmark. This structural variation is deliberate: it tests models across low- and high-channel regimes, short- and long-horizon reasoning, and all three modality alignment types, from homogeneous synchronized signals like PTB-XL and Chapman-Shaoxing to heterogeneous unaligned mixtures of time series and static embeddings in MIMIC-IV and CirCor. Full per-dataset statistics are in Appendix C.

4 Experiments

4.1 Experimental Setup

This section presents the experimental results and analysis. We evaluate model robustness against incomplete data across diverse settings, guided by three questions: **(a) Datasets:** Do models show consistent robustness across dataset types and clinical domains? **(b) Models:** Can different fusion architectures maintain robustness under missing data? **(c) Missingness:** Do models respond differently to modality missing versus within-modality missing? The datasets selected and model configurations used throughout this work are described as follows.

Datasets: We use 9 datasets covering the domains and modality types defined in Section 3. These include medical data (MIMIC-IV, Challenge-2012), heart signals (PTB-XL, Chapman-Shaoxing), cardiac auscultation (CirCor), brain signals (Sleep-EDF), wearable data (PPG-DaLiA, WESAD), and activity tracking (HAR-UP). Categorized by modality structure, these datasets span three types: Type 1 (homogeneous and aligned), Type 2 (heterogeneous and aligned), and Type 3 (heterogeneous and unaligned), as detailed in Appendix C. Table 2 lists the specific details for all 9 datasets.

Models: We evaluate 6 representative multimodal fusion architectures spanning multiple design paradigms (Figure 1, right): channel-independent models (CLIMB [8], MIRA [23]), a shared-specific decomposition model (ShaSpec [43]), and MoE-fusion models (Flex-MoE [49], FuseMoE [13], Mae-

Table 4: Clean data results on nine datasets, averaged over three independent seeds. **AUC**: AUROC; **F1**: Macro-F1. Bold denotes the best result per dataset.

Dataset	CLIMB		MIRA		Flex-MoE		ShaSpec		FuseMoE		Maestro	
	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1
HAR-UP	.989	.919	.976	.878	.855	.780	.914	.885	.937	.825	.991	.944
PTB-XL	.838	.545	.830	.496	.772	.466	.865	.635	.677	.210	.883	.621
Chapman	.786	.329	.827	.388	.727	.363	.850	.502	.669	.331	.776	.428
Sleep-EDF	.980	.728	.949	.608	.973	.701	.984	.763	.932	.586	.971	.698
PPG-DaLiA	.942	.667	.937	.668	.930	.631	.959	.735	.927	.627	.956	.762
WESAD	.701	.501	.685	.470	.684	.434	.674	.451	.583	.369	.736	.494
MIMIC-IV	.782	.411	.741	.371	.760	.403	.656	.311	.811	.420	.805	.393
Challenge	.697	.322	.742	.383	.767	.393	.610	.122	.806	.423	.812	.428
CirCor	.649	.447	.601	.387	.567	.357	.678	.444	.655	.409	.796	.573

stro [29]). Table 3 summarizes their architectures and key fusion strategies; complete implementation details and hyperparameter settings are in Appendix B.

Missing Data and Evaluation Protocol: We evaluate two missing-data conditions (Figure 2). Under **modality missing**, each channel is independently dropped with probability $p \in \{0.2, 0.5\}$, simulating whole-sensor failures; at least one channel is always retained. Under **within-modality missing**, non-overlapping contiguous blocks (5–10% of T each) are masked per channel to a total fraction $b \in \{0.2, 0.5\}$, simulating transient signal interruptions; static-feature channels such as CirCor demographics are excluded. All patterns are generated deterministically from a (seed, sample_id) pair and applied uniformly to train, validation, and test splits, so every model receives identical masks. Maestro is the sole exception, using curriculum modality dropout during training while evaluating under the same protocol. Note that modality missing operates at channel level rather than strict modality level; see Appendix D.4.1. Full details are in Appendix D.

4.2 Main Results

We evaluate six models on nine datasets. We report AUROC and Macro-F1 under three conditions: clean data, modality missing, and within-modality missing. The tables report these as AUC and F1 columns for each model (AUC: AUROC; F1: Macro-F1).

Clean Data Results. Table 4 shows clean-data baselines. No single model dominates: ShaSpec leads on Sleep-EDF, PPG-DaLiA, and Chapman; Maestro on PTB-XL, WESAD, HAR-UP, Challenge-2012, and CirCor; FuseMoE leads on MIMIC-IV (0.811 AUROC). These clean-data scores serve as the no-missing baseline shared by both modality-missing and within-modality-missing evaluations.

Modality Missing Results. Table 5 shows results at 20% and 50% drop rates. Channel-independent models show the strongest robustness: on Chapman at 50% drop, CLIMB retains $\Delta\text{AUROC} = -0.014$ and MIRA loses only $\Delta\text{AUROC} = -0.017$. This reflects their design of encoding each channel independently so that dropped modalities do not corrupt remaining representations. Maestro, trained with curriculum dropout, also maintains competitive performance across most datasets. In contrast, fusion-centric architectures suffer more: FuseMoE loses 0.075 AUROC on PTB-XL at 50% drop, and ShaSpec shows instability on Challenge-2012 where Macro-F1 collapses to 0.000 at 20% drop. This suggests that shared-parameter fusion models are more sensitive to the complete absence of expected input streams.

Within-Modality Missing Results. Table 6 shows within-modality missing results. ShaSpec is most robust on ECG datasets: PTB-XL AUROC drops only $\Delta = -0.007$ at the 50% block rate, likely because masked temporal segments in ECG signals preserve enough morphological context for classification. Flex-MoE maintains stability on MIMIC-IV across both block rates ($\Delta\text{AUROC} \leq 0.010$), suggesting that its mixture-of-experts routing can compensate for localized temporal gaps. However, Maestro shows large F1 degradation on Sleep-EDF at block 50% (F1: .443 vs. .698 clean, $\Delta = -0.255$), indicating that curriculum-dropout-based robustness does not fully transfer to within-modality missing.

Table 5: Modality missing results on nine datasets at 20% and 50% drop rates, averaged over three independent seeds. **AUC**: AUROC; **F1**: Macro-F1. Bold denotes the best result per dataset per rate.

Dataset	Rate	CLIMB		MIRA		Flex-MoE		ShaSpec		FuseMoE		Maestro	
		AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1
HAR-UP	20%	.985	.932	.966	.888	.811	.720	.907	.887	.882	.789	.987	.935
	50%	.978	.893	.936	.855	.752	.676	.887	.865	.780	.696	.962	.878
PTB-XL	20%	.822	.512	.825	.498	.737	.452	.860	.630	.649	.270	.873	.611
	50%	.803	.496	.813	.485	.707	.461	.846	.614	.602	.233	.848	.567
Chapman	20%	.783	.324	.831	.414	.711	.348	.849	.502	.638	.337	.763	.415
	50%	.772	.314	.810	.397	.681	.345	.844	.505	.580	.291	.726	.391
Sleep-EDF	20%	.973	.688	.926	.560	.962	.654	.980	.737	.890	.505	.952	.645
	50%	.947	.607	.870	.456	.923	.567	.960	.663	.797	.383	.889	.523
PPG-DaLiA	20%	.930	.645	.933	.663	.909	.570	.953	.712	.901	.562	.931	.675
	50%	.904	.553	.912	.589	.876	.479	.936	.645	.870	.455	.864	.501
WESAD	20%	.666	.454	.637	.429	.682	.431	.642	.434	.567	.360	.721	.474
	50%	.649	.444	.620	.427	.675	.415	.651	.448	.592	.374	.696	.471
MIMIC-IV	20%	.750	.361	.725	.364	.737	.376	.638	.298	.777	.385	.777	.395
	50%	.708	.340	.681	.338	.696	.341	.615	.272	.736	.346	.729	.360
Challenge	20%	.640	.266	.724	.322	.752	.383	.645	.000	.692	.331	.781	.392
	50%	.667	.314	.697	.298	.717	.347	.607	.168	.616	.257	.714	.312
CirCor	20%	.652	.428	.596	.395	.529	.315	.659	.443	.599	.387	.783	.561
	50%	.645	.433	.560	.373	.539	.337	.677	.438	.606	.409	.755	.508

Table 6: Within-modality missing results on nine datasets at 20% and 50% block rates, averaged over three independent seeds. **AUC**: AUROC; **F1**: Macro-F1. Bold denotes the best result per dataset per rate.

Dataset	Rate	CLIMB		MIRA		Flex-MoE		ShaSpec		FuseMoE		Maestro	
		AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1
HAR-UP	20%	.983	.919	.969	.901	.826	.742	.883	.851	.866	.782	.991	.941
	50%	.940	.843	.937	.830	.795	.710	.866	.820	.842	.756	.971	.802
PTB-XL	20%	.822	.519	.836	.521	.736	.429	.863	.634	.662	.236	.877	.620
	50%	.798	.457	.818	.478	.706	.404	.858	.630	.608	.158	.856	.514
Chapman	20%	.777	.313	.814	.400	.706	.343	.842	.493	.634	.323	.767	.369
	50%	.741	.246	.809	.371	.669	.322	.835	.487	.599	.298	.730	.295
Sleep-EDF	20%	.979	.733	.942	.594	.967	.675	.983	.747	.914	.555	.956	.625
	50%	.973	.698	.927	.568	.961	.650	.981	.740	.887	.505	.893	.443
PPG-DaLiA	20%	.934	.641	.936	.671	.918	.599	.950	.688	.920	.604	.943	.701
	50%	.907	.573	.934	.651	.905	.555	.939	.620	.914	.567	.882	.467
WESAD	20%	.696	.483	.664	.454	.699	.439	.659	.429	.589	.374	.726	.460
	50%	.656	.448	.622	.411	.681	.412	.639	.415	.586	.362	.683	.413
MIMIC-IV	20%	.774	.404	.757	.390	.759	.398	.653	.309	.805	.417	.798	.407
	50%	.752	.379	.744	.357	.750	.383	.645	.309	.794	.374	.796	.412
Challenge	20%	.680	.299	.729	.341	.757	.387	.638	.006	.800	.393	.814	.433
	50%	.674	.312	.715	.357	.742	.355	.628	.138	.793	.413	.790	.393
CirCor	20%	.659	.442	.625	.437	.571	.344	.646	.389	.630	.399	.774	.548
	50%	.632	.430	.565	.367	.572	.352	.663	.368	.623	.397	.759	.477

4.3 Insights

From the benchmark results across nine datasets and six fusion architectures, we identify three consistent trends. All trends are stable across the three independent seeds; per-seed results are in Appendix E.1.

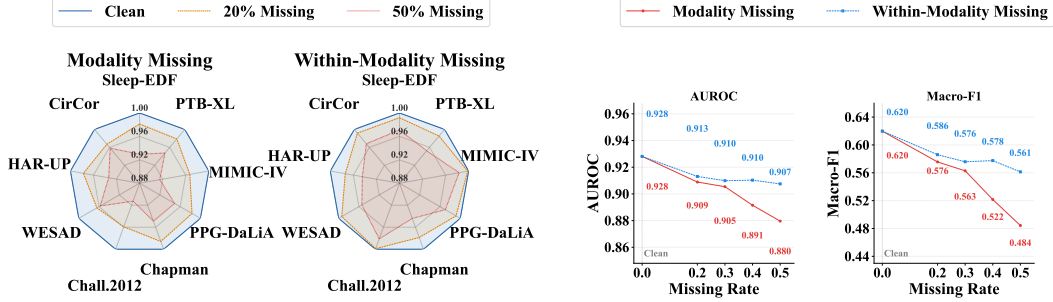


Figure 3: Degradation analysis. **left:** Radar chart of AUROC drop ($\Delta\text{AUROC} = \text{clean} - \text{missing}$, averaged over three seeds) across all 9 datasets under modality and within-modality missing at 20% and 50% rates; larger area indicates greater overall sensitivity, and each axis corresponds to one dataset. **right:** Detailed degradation trajectory of Flex-MoE on PPG-DaLiA: both AUROC and Macro-F1 decline steeply as missing rate increases, with modality missing inducing sharper drops than within-modality missing.

Architecture family predicts robustness better than parameter count. Channel-independent models (CLIMB, MIRA) are the most stable under modality missing, as isolated channel encoding prevents cross-channel interference. Shared-specific models handle within-modality missing best on homogeneous correlated signals (PTB-XL $\Delta\text{Block50\%} = -0.006$). FuseMoE ($\approx 256\text{M}$ parameters) is consistently among the most vulnerable models under both conditions, confirming that scale does not substitute for structural robustness design. Per-family degradation is detailed in Appendix E.2.

Dataset structure determines which missing type causes more harm. Datasets with many channels and short sequences (e.g., Challenge-2012: $C = 42$, $T = 48$) suffer more from modality missing. Each channel carries dense information, while a missing block covers only 2 to 5 time steps and contributes little to overall loss. In contrast, long-sequence datasets (e.g., Sleep-EDF: $T = 3000$) are also vulnerable to within-modality missing, as models relying on temporal continuity are more affected when a contiguous segment is lost. Per-dataset breakdowns are in Appendix E.

Curriculum dropout protection is bounded by its training configuration. Models using curriculum modality dropout (Appendix D.5), such as Maestro, show substantially lower degradation under modality missing (Challenge-2012 $\Delta\text{Mod50\%} = -0.095$ vs. -0.172 for FuseMoE). However, this protection only holds within the maximum dropout rate seen during training. On PPG-DaLiA (5 modalities), a 50% modality missing rate removes 2.5 modalities on average (in this case, 3 modalities would be removed), which exceeds Maestro’s training upper bound of $\approx 40\%$ (at most 2 modalities). As a result, Maestro’s degradation becomes the worst in the dataset ($\Delta = -0.099$). The curriculum upper bound must cover the worst expected missing rate at test time.

4.4 Degradation Analysis

Figure 3(left) summarizes AUROC degradation across all datasets and models, showing that degradation varies by dataset structure and architecture family (see Tables 19–20). Figure 3(right) examines Flex-MoE on PPG-DaLiA as a representative case study. Macro-F1 drops more sharply than AUROC under both conditions (modality 50%: $\Delta\text{F1} = -0.152$ vs. $\Delta\text{AUROC} = -0.054$), and modality missing causes a steeper decline than within-modality missing ($\Delta\text{F1} = -0.076$ at within-modality 50%). The F1–AUROC gap arises because AUROC measures ranking ability and is relatively insensitive to shifts in predicted score magnitudes, whereas F1 directly reflects classification decision quality; when expert routing loses an entire input modality, the model tends toward majority-class predictions and F1 collapses faster. The sharper drop under modality missing reflects the complementary nature of PPG-DaLiA’s modalities: removing an entire channel (e.g., PPG, skin temperature, or accelerometer) deprives the MoE gating mechanism of one expert’s input entirely, while within-modality missing only corrupts a local temporal window and leaves the gating structure intact. The degradation trajectory also steepens nonlinearly from 20% to 50% under modality missing, whereas within-modality missing degrades more gradually, suggesting that MoE-fusion architectures are disproportionately sensitive to complete channel loss as missing rate increases.

Table 7: Effect of diffusion-based imputation on PTB-XL under modality missing. **Raw**: models trained on zero-filled dropped-modality input. **Diffusion**: same model trained on diffusion-reconstructed input. Δ denotes the performance change from Raw to Diffusion Imputed. **AUC**: AUROC; **F1**: Macro-F1. Three seeds; mean $\pm$ std.

Model	Rate	Raw		Diffusion Imputed		Δ	
		AUC	F1	AUC	F1	AUC	F1
CLIMB	20%	.822 $\pm$.009	.512 $\pm$.012	.816 $\pm$.001	.503 $\pm$.008	-.006	-.009
	50%	.803 $\pm$.006	.496 $\pm$.024	.796 $\pm$.006	.448 $\pm$.018	-.007	-.048
Flex-MoE	20%	.737 $\pm$.019	.452 $\pm$.030	.730 $\pm$.011	.405 $\pm$.015	-.007	-.046
	50%	.707 $\pm$.007	.461 $\pm$.005	.678 $\pm$.009	.341 $\pm$.016	-.028	-.120
ShaSpec	20%	.860 $\pm$.003	.630 $\pm$.004	.851 $\pm$.001	.622 $\pm$.002	-.008	-.008
	50%	.846 $\pm$.004	.614 $\pm$.002	.825 $\pm$.002	.598 $\pm$.001	-.021	-.015

Table 8: Effect of diffusion-based imputation on PTB-XL under within-modality missing. **Raw**: models trained on zero-filled corrupted input. **Diffusion**: models trained on diffusion-reconstructed input. Δ denotes the performance change from Raw to Diffusion Imputed. **AUC**: AUROC; **F1**: Macro-F1. Three seeds; mean $\pm$ std.

Model	Rate	Raw		Diffusion Imputed		Δ	
		AUC	F1	AUC	F1	AUC	F1
CLIMB	20%	.822 $\pm$.003	.519 $\pm$.015	.826 $\pm$.003	.526 $\pm$.003	+.004	+.007
	50%	.798 $\pm$.004	.457 $\pm$.015	.812 $\pm$.005	.487 $\pm$.006	+.014	+.030
Flex-MoE	20%	.736 $\pm$.012	.429 $\pm$.025	.760 $\pm$.009	.435 $\pm$.026	+.024	+.006
	50%	.706 $\pm$.007	.404 $\pm$.015	.744 $\pm$.004	.427 $\pm$.013	+.038	+.023
ShaSpec	20%	.863 $\pm$.004	.634 $\pm$.006	.867 $\pm$.001	.636 $\pm$.001	+.004	+.003
	50%	.858 $\pm$.004	.630 $\pm$.005	.860 $\pm$.001	.631 $\pm$.001	+.001	+.001

4.5 Imputation Effect

We evaluate an impute-then-predict strategy on PTB-XL. A conditional score-based diffusion model [14, 39] is trained at native ECG resolution ($T=1000$, 50 diffusion steps) and conditioned on observed channels to reconstruct missing segments offline; each downstream classifier then trains on the pre-imputed signals (Appendix F.1). Results are reported in Tables 7–8. Under modality missing, the strategy offers no benefit: AUROC drops by 0.004 to 0.028 across all settings, and Macro-F1 degrades more sharply. Without cross-modal conditioning guidance, reconstructing a fully absent modality is out of distribution and yields unreliable estimates. Models on zero-filled inputs learn to isolate missingness, so low-quality reconstructions instead corrupt the representation space. Under within-modality missing, imputation improves all three models on PTB-XL. Gains scale with the missing rate: Flex-MoE recovers the most (+0.038 AUROC at 50%), as its expert routing is most sensitive to corrupted inputs. CLIMB improves moderately (+0.014 AUROC at 50%). ShaSpec benefits least, gaining only +0.001 AUROC at 50%, because its shared-specific decomposition already handles within-lead gaps. These gains do not close the gap to clean performance, so imputation serves as a partial fix rather than a complete solution. These divergent outcomes suggest applying imputation selectively. Whether these patterns hold beyond PTB-XL remains open.

5 Conclusion

We introduce MuteBench, a benchmark evaluating multimodal fusion architectures under two missing-data conditions across 9 clinical datasets, 6 models, and 810 experimental runs. Architecture family predicts robustness better than parameter count: channel-independent models handle modality missing well, while shared-specific models excel at within-modality missing on homogeneous signals but degrade on short or imbalanced sequences. A PTB-XL case study suggests that diffusion imputation can improve within-modality missing recovery but provides little benefit when an entire channel is absent. Validating this finding broadly across more diverse datasets and settings remains important future work. We recommend matching architecture choice to the dominant failure mode and setting curriculum dropout bounds to the worst expected missing rate.

Our findings also suggest several directions for future work: designing architectures that balance shared and modality-specific representations across heterogeneous signals, developing fusion mechanisms that degrade gracefully without explicit dropout curricula, and extending diffusion-based imputation beyond a single dataset. By releasing our complete benchmark suite, missingness library, and pretrained checkpoints, we hope MuteBench provides the practical guidance that real-world practitioners need and establishes a reproducible and extensible foundation for developing the next generation of robust multimodal fusion models for clinical AI.

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A Broader Impacts

MuteBench provides practitioners with concrete, dataset-aware guidance for selecting multimodal fusion architectures that remain reliable under realistic sensor failures. Making these failure modes explicit and quantifiable reduces the risk of deploying clinical AI systems that degrade silently, which could otherwise lead to missed diagnoses or delayed alerts in ICU monitoring, cardiac screening, and wearable health applications. The released code and missingness library also lower the barrier for other researchers to conduct robustness evaluations, encouraging more systematic stress-testing in clinical ML.

The benchmark characterizes the conditions under which each architecture is most vulnerable, which in principle could be used to identify operating regimes where clinical AI systems are most susceptible to failure. However, all evaluated datasets and model codebases are already publicly available, so the benchmark does not introduce new attack surfaces. We do not foresee direct paths to harmful misuse.

This section provides detailed technical descriptions of the six representative multimodal fusion architectures evaluated in our main experiments (Section 4.1). Each model is cloned from its official repository and adapted to a unified multi-dataset evaluation protocol. Below, we expand on the architectural backbone, fusion strategy, and specific mechanisms for handling missing modalities for each baseline.

B Detailed Baseline Model Architectures

All six models are adopted from publicly available or previously published work and evaluated purely as baselines; none is a contribution of this paper. Each model is cloned from its official repository (or internal codebase in the case of Maestro, which is cited as a prior work) and adapted to our unified multi-dataset evaluation protocol with minimal modification: we replace the original classification head with a task-appropriate output layer and apply the shared missingness injection interface described in Appendix D.3. Hyperparameter selection follows the published recommendations for each model; no model receives preferential tuning. Table 9 summarises the origin, code availability, pretraining status, and approximate parameter count for each baseline.

Table 9: Summary of baseline model provenance and adaptation. All models are evaluated without pretraining on external data. Parameter counts are approximate and dataset-independent (i.e., excluding the final classification head). Adaptation level: *minimal* = head replacement only; *moderate* = head replacement plus task-specific loss.

Model	Family	Original Venue	Official Code	Pretraining	Approx. Params
CLIMB	Chan.-Indep.	Published [8]	Yes	No	~9.5M
MIRA	Chan.-Indep.	Published [23]	Yes	No	~30M
ShaSpec	Shared-Spec.	Published [43]	Yes	No	~2.3M
Flex-MoE	MoE-Fusion	Published [49]	Yes	No	~1.5M
FuseMoE	MoE-Fusion	Published [13]	Yes	No	~5M
Maestro	MoE-Fusion	Published [29]	Yes	No	~1M

B.1 CLIMB

CLIMB (Clinical Large-scale Integrative Multi-modal Benchmark) is built on the BenchMD framework [46]. It is designed as a modality-agnostic benchmark framework covering diverse medical data types under a single evaluation suite.

Architecture.

- **Backbone:** A Domain Agnostic Transformer (DAT) with embedding dimension 256, depth of 12 Transformer layers, 8 attention heads, and MLP dimension 512.
- **Channel-independent encoding:** Each time-series channel is independently projected to a 256-dimensional embedding space. Channels never interact during encoding; only the aggregation step combines them.
- **Aggregation:** Mean pooling across all channel embeddings produces a single fixed-size representation, which is passed to the classification head.

Missing modality handling. CLIMB does not implement explicit modality dropout. Its channel-independent encoding provides implicit robustness: absent channels are zeroed at the input level, and mean pooling naturally down-weights missing inputs without requiring training-time dropout.

B.2 Flex-MoE

Flex-MoE [49] is a sparse Mixture-of-Experts framework specifically designed to handle arbitrary combinations of missing modalities at inference time (NeurIPS 2024 Spotlight). It builds combinatorial robustness directly into the MoE routing mechanism.

Architecture.

- **Backbone:** Standard Transformer encoder layers; the MoE module [32] replaces the feed-forward sub-layer.
- **Missing Modality Bank:** For each observed modality subset, a learned embedding bridges the gap between the observed combination and the full-modality representation, preventing the model from collapsing to any single dominant modality.
- **G-Router (Generalized Router):** Trained exclusively on complete-modality samples to inject generalized cross-modal knowledge into the expert pool, ensuring experts learn coherent cross-modal representations before specialization.
- **S-Router (Specialized Router):** Uses top-1 hard routing to assign each incomplete modality combination to the single most appropriate expert. The expert index is determined by the observed modality combination index, providing a deterministic and interpretable routing scheme.

Missing modality handling. The Missing Modality Bank explicitly models every possible subset of observed modalities, making Flex-MoE inherently robust to any combination of missing inputs without additional training-time dropout.

B.3 MIRA

MIRA [23] is a large-scale foundation model for medical time series, pretrained on 454 billion time points from heterogeneous clinical sources (ICU waveforms, hospital EHR, and epidemiological signals; NeurIPS 2025). Its design priority is zero-shot generalization across datasets with different sampling rates, signal types, and temporal horizons.

Architecture.

- **Continuous-Time Rotary Positional Encoding (CT-RoPE):** Extends standard RoPE to encode exact continuous-valued timestamps rather than integer positions, allowing a single model to natively process signals sampled at 250 Hz (ECG), 32 Hz (wearables), once per minute (vitals), and once per day (lab values) within the same attention computation.
- **Frequency-Specialized Mixture-of-Experts:** Each expert specializes in a different temporal frequency regime (e.g., high-frequency cardiac waveforms vs. low-frequency clinical trends). Frequency-aware routing ensures physiologically distinct signal types are processed by appropriate expert sub-networks rather than competing for the same representation space.

- **Neural ODE extrapolation:** After encoding the observed time points, a Neural ODE (via `torchdiffeq`) models latent dynamics continuously, enabling inference at arbitrary timestamps without requiring a fixed-horizon output head. For classification fine-tuning, the continuous forecasting head is replaced with a task-specific classification head.

Missing modality handling. MIRA uses a mask field in its data format to indicate missing time points. Masked positions are excluded from attention computation via attention masking, so partial-modality inputs are processed without architectural changes.

B.4 ShaSpec

ShaSpec [43] was originally designed for 3D MRI brain tumor segmentation under missing-modality conditions (CVPR 2023). Its core contribution is an explicit decomposition of features into shared and modality-specific components. In our benchmark, we adapt this framework to 1D temporal signals.

Architecture.

- **Shared encoder:** A stack of lightweight 1D convolutional blocks (`ConvBlock1d`) learns modality-agnostic temporal representations using `InstanceNorm1d`. The ResNet-50 backbone and ASPP module from the original 3D model are not carried over.
- **Modality-specific encoders:** Each modality has its own `Encoder1D` branch (identical structure to the shared encoder) that captures modality-exclusive temporal features.
- **Shared-specific fusion (`CompositionalLayer1D`):** Shared and specific features are fused via a residual connection: $\text{fused} = \text{shared} + \text{conv}(\text{cat}(\text{shared}, \text{specific}))$, preserving the core ShaSpec design principle.
- **Classification head:** Global average pooling over the fused temporal features followed by a linear layer.

Missing modality handling. Missing modalities are handled by zeroing out the corresponding input channels. The modality-specific encoder for a missing input still runs on the zero input; the model falls back to the shared encoder features from available modalities when constructing the fused representation. No explicit modality dropout warmup is implemented.

B.5 FuseMoE

FuseMoE is a multimodal time-series fusion model built on a cross-modal Transformer with Sparse Mixture-of-Experts feed-forward layers [13]. Its design targets irregularly sampled and heterogeneous physiological signals by combining cross-attention fusion with learned expert specialization.

Architecture.

- **Per-modality projection:** Each modality is independently projected to a shared embedding space via a `Conv1d` layer, preserving local temporal structure before cross-modal interaction.
- **Token type embeddings:** A learned embedding is added to each token to identify its source modality, providing the cross-encoder with explicit modality identity information.
- **TransformerCrossEncoder with Sparse MoE:** The fusion module uses multi-head cross-attention where query tokens attend to key-value pairs from all modalities. The feed-forward sub-layer is replaced by a Sparse MoE block [32] with top- k gating ($k = 1, 4$ experts), allowing different experts to specialize in different inter-modality interaction patterns.
- **mTAND (Multi-Time Attention)** A `multiTimeAttention` module handles irregular sampling by learning continuous-time attention weights, enabling the model to process asynchronous sensor streams.

Model configuration: `embed_dim=128, num_heads=8, num_layers=2, num_experts=4, top_k=1`.

Missing modality handling. Missing-modality tokens are replaced with learned zero-padded embeddings; an attention mask prevents the model from attending to missing-modality positions in the cross-encoder.

B.6 Maestro

Maestro [29] is a previously published multi-dataset medical time-series classifier included here solely as a representative MoE-fusion baseline. It is not a contribution of this paper, and its hyperparameters and training protocol are held to the same standard of transparency as all other baselines. Its stronger clean-data performance on several datasets reflects its architectural design choices, not preferential treatment in our evaluation.

Maestro is built on a custom multi-stage sparse architecture built on a custom multi-stage sparse architecture. It does not rely on any standard backbone (e.g., Transformer, Informer, ResNet); instead, it combines SAX symbolic tokenization, per-modality sparse attention encoding, cross-modal sparse attention, and loss-free Sparse MoE routing [32]. It is designed to scale efficiently to long physiological sequences while maintaining per-modality specialization.

Architecture.

- **SAX Symbolic Tokenization:** Each modality’s time series is compressed into a symbolic sequence via Symbolic Aggregate approXimation (SAX). A dedicated reserved symbol represents missing modalities, enabling the model to natively distinguish absent from present inputs at the token level.
- **ModalityPositionalEncoder:** Combines sinusoidal temporal positional encodings with learned modality embeddings (one per sensor/channel type), giving each token a two-part identity: *when* it occurred and *which modality* it belongs to.
- **Per-modality Sparse Attention Encoder:** Each modality is encoded independently by a sparse self-attention module whose attention budget is dynamically controlled by a modality-aware gate, followed by 1D convolution and max-pooling distillation for sequence compression. This reduces per-modality self-attention complexity from $O(L^2)$ to $O(L \log L)$.
- **Cross-modal Sparse Attention:** Token sequences from all modalities are concatenated and processed by a sparse cross-modal multi-head attention layer, enabling inter-modality information exchange at $O(L \log L)$ complexity.
- **Sparse MoE Routing:** A top- k Sparse MoE block (4 experts) replaces the standard feed-forward layer. Experts automatically specialize by modality combination without auxiliary balancing losses (loss-free MoE).
- **CrossAttnTransformerClf:** The top-level classifier uses cross-attention between a learned class token and the encoded multi-modality token sequence, followed by a linear classification head.

Model configuration: d_model=32, nhead=16, num_layers=2, num_experts=4.

Missing modality handling. Maestro applies two complementary mechanisms for missing modality handling. At inference time, missing modality streams are replaced with a learned reserved token produced by SAX symbolic tokenization, allowing the model to natively accept any combination of present and absent modalities without architectural changes. During training, curriculum modality dropout [4] linearly increases the per-modality dropout probability from 0 to a maximum of 40% over a warmup schedule of 10 epochs, after which it remains fixed. Dropped modality streams are zeroed out and excluded from attention computation via an attention mask. This structured exposure to progressively harder missing-data scenarios is the primary source of Maestro’s robustness advantage over other MoE-fusion models at test time.

C Detailed Dataset Specifications

We provide full specifications for all nine datasets below, organized by modality type. For each dataset we describe its clinical significance, label semantics, channel configuration, dataset scale, and the preprocessing steps applied before training. The primary evaluation metric is Macro-AUROC for all datasets, with Macro-F1 reported alongside.

C.1 Dataset Selection Criteria

We use three selection rules to create a rigorous testbed that measures both spatial and temporal robustness:

- **Domain Diversity:** We include multiple fields (intensive care, heart signals, activity tracking) to show that a model’s spatial compensation ability holds across different physical signals.
- **Data Shape:** We select datasets with various channel counts and sequence lengths to test the limits of temporal reasoning. Datasets with many channels evaluate spatial redundancy, while those with long time steps measure how well models interpolate missing blocks.
- **Task Types:** We include binary, multi-class, and multi-label tasks to verify that our conclusions hold true regardless of the clinical objective.

C.2 Dataset Overall Statistics

The benchmark covers 9 datasets from 7 clinical and physiological domains, totalling over 125,000 samples. Table 2 summarizes their key properties.

Scale and Task Diversity. Dataset sizes range from 515 samples (HAR-UP) to 64,700 samples (PPG-DaLiA). The benchmark covers three task formats: binary classification (HAR-UP, MIMIC-IV, Challenge-2012), multi-class classification with 3 to 8 categories (Sleep-EDF, PPG-DaLiA, WESAD, CirCor), and multi-label prediction (PTB-XL, Chapman-Shaoxing). All datasets are evaluated with Macro-AUROC as the primary metric, with Macro-F1 reported alongside.

Channel and Sequence Diversity. Channel counts range from 2 (WESAD) to 78 (CirCor), providing a testbed for both low-dimensional and high-dimensional fusion. Sequence lengths range from 48 time steps (MIMIC-IV, Challenge-2012) to 3000 time steps (Sleep-EDF). Sampling rates span from once-per-hour clinical observations to 4,000 Hz physiological waveforms. This variation deliberately stresses models across different spatial redundancy levels and temporal resolutions, ensuring that benchmark conclusions generalize beyond any single signal regime.

C.3 Formal Modality Type Definitions

We categorize the data into three types to isolate how structural alignment affects a model’s ability to handle missing information:

Type 1: Homogeneous and Aligned Datasets. All modalities in this group share the same physical signal format (inertial motion or ECG voltage) and are recorded synchronously along a common time axis.

- **HAR-UP (UP-Fall Detection Dataset).** HAR-UP[27] is a multimodal fall-detection dataset from the Autonomous University of Puebla (Mexico), designed for ambient-assisted living applications. Falls are one of the leading causes of injury-related mortality in the elderly, and automated detection from body-worn sensors can trigger emergency alerts within seconds. The dataset covers controlled fall events and typical daily living activities across 17 subjects.

The task is **binary classification**. Label 1 (Fall) covers forward, backward, and lateral fall events caused by tripping, stumbling, or loss of balance. Label 0 (Non-Fall) covers activities of daily living including walking, sitting, standing, and postural transitions.

Five inertial measurement unit (IMU) sensors are placed at five body positions: ankle, right pocket, belt, necklace, and wrist. Each sensor records a 3-axis accelerometer signal and a 3-axis gyroscope signal, yielding $5 \times 6 = 30$ channels in total. Because all channels share the same physical signal type (IMU motion), this is a **Type 1 (homogeneous and aligned)** dataset.

After sliding-window segmentation the dataset contains **515 samples**. Each sample has shape $\mathbb{R}^{30 \times 140}$ ($C = 30, T = 140$), corresponding to approximately 2.8 seconds at ≈ 50 Hz.

We use all 30 sensor channels without resampling. A fixed sliding window of 140 time steps is applied to the raw sensor streams. Z-score normalization is applied to each channel independently. Data splits are stratified by subject to prevent data leakage.

- **PTB-XL.** PTB-XL[42] is the largest publicly available annotated clinical 12-lead ECG dataset, released by the Physikalisch-Technische Bundesanstalt (PTB) in Berlin. It covers the full diagnostic

spectrum encountered in cardiology practice and serves as the de facto standard benchmark for ECG multi-label classification [38]. Automated ECG interpretation has direct clinical value in screening large populations and supporting cardiologists in high-volume environments.

The task is **multi-label classification** across five diagnostic superclasses derived from the 71 SCP-ECG codes in the original annotation. A patient may simultaneously carry multiple diagnoses:

- **NORM:** Normal ECG, no pathological finding identified.
- **MI (Myocardial Infarction):** ST-segment elevation, pathological Q-waves, or T-wave inversion consistent with current or prior infarction.
- **STTC (ST/T-wave Change):** Non-specific repolarization abnormalities including ST depression or elevation not meeting STEMI criteria, and diffuse T-wave changes.
- **CD (Conduction Disturbance):** Bundle branch blocks (LBBB/RBBB), fascicular blocks, Wolff-Parkinson-White syndrome, and first-/second-/third-degree AV conduction delays.
- **HYP (Hypertrophy):** Left or right ventricular hypertrophy and left or right atrial enlargement.

The standard clinical 12-lead ECG configuration is used: limb leads I, II, III, aVR, aVL, aVF, and precordial leads V1–V6 ($C = 12$). All leads are derived from body-surface electrodes and are synchronously sampled, making this a **Type 1 (homogeneous and aligned)** dataset.

The dataset contains **21,837 recordings** from 18,885 patients. After preprocessing each sample has shape $\mathbb{R}^{12 \times 250}$ ($C = 12$, $T = 250$).

Original signals are recorded at 500 Hz (5000 time steps per 10-second recording). We resize the sequence to $T = 250$ via linear interpolation to align input shapes across all model architectures. Z-score normalization is applied per lead. We use the official 10-fold stratified cross-validation split provided with the dataset.

- **Chapman-Shaoxing.** The Chapman-Shaoxing ECG dataset[52] is a large-scale 12-lead ECG corpus jointly collected by Chapman University (USA) and Shaoxing People’s Hospital (China). It is designed specifically for cardiac arrhythmia and conduction-abnormality classification and provides a realistic distribution of rhythm types from outpatient and inpatient clinical settings. Automated arrhythmia detection reduces missed diagnoses and enables earlier intervention for life-threatening conditions such as atrial fibrillation and STEMI.

The task is **multi-label classification** with 7 diagnostic label groups derived by mapping SNOMED CT codes via `generate_labels.py`; a recording may receive multiple simultaneous labels (47% of recordings carry ≥ 2 labels):

- **Normal:** Sinus bradycardia, normal sinus rhythm, or sinus tachycardia.
- **CD (Conduction Disturbance):** First-degree AV block, complete left/right bundle branch block, Wolff–Parkinson–White, and related conduction delays.
- **MI (Myocardial Infarction):** ST-elevation patterns consistent with current or prior myocardial infarction.
- **STTC (ST/T-wave Change):** ST-segment depression or elevation and T-wave changes not meeting MI criteria.
- **Other:** Remaining arrhythmia codes not captured by the above groups (e.g., atrial bigeminy, aberrant ventricular conduction, early repolarisation).
- **AFib (Atrial Fibrillation):** Atrial fibrillation and atrial flutter.
- **HYP (Hypertrophy):** Left or right ventricular hypertrophy and atrial enlargement.

All 12 ECG leads are used ($C = 12$) and recorded synchronously, making this a **Type 1 (homogeneous and aligned)** dataset.

The dataset contains **10,646 recordings**. After preprocessing each sample has shape $\mathbb{R}^{12 \times 1000}$ ($C = 12$, $T = 1000$).

Original 500 Hz signals are downsampled to 100 Hz using anti-aliasing filtering, yielding $T = 1000$ time steps per 10-second recording. Downsampling suppresses high-frequency noise and reduces overfitting. Z-score normalization is applied per lead.

Type 2: Heterogeneous and Aligned Datasets. In this group, distinct physiological signal types are recorded simultaneously at a shared (or unified) sampling rate. After resampling all channels share a common time axis, but they originate from fundamentally different physiological processes.

- **Sleep-EDF.** The Sleep-EDF dataset[21] is a publicly available polysomnography (PSG) corpus from PhysioNet [12], originating from a prospective study of sleep and ageing in healthy subjects.

Automated sleep-stage scoring is clinically critical for diagnosing sleep disorders such as insomnia, obstructive sleep apnoea, and narcolepsy. Manual PSG scoring by trained technicians is time-consuming, expensive, and subject to inter-rater variability.

The task is **five-class sleep-stage classification** following the AASM scoring rules. Each 30-second epoch receives exactly one label:

- **W (Wake):** Subject is awake; EEG shows high-frequency, low-amplitude activity and voluntary eye movements.
- **N1 (NREM Stage 1):** Sleep onset; theta waves (4–7 Hz) dominate the EEG, muscle tone decreases, and slow eye movements appear.
- **N2 (NREM Stage 2):** Established light sleep; defined by K-complexes and sleep spindles (12–15 Hz bursts) in the EEG.
- **N3 (NREM Stage 3 / Slow-Wave Sleep):** Deep sleep; high-amplitude delta waves (<2 Hz, $>75 \mu\text{V}$) occupy $\geq 20\%$ of the epoch.
- **R (REM):** Rapid Eye Movement sleep; EEG resembles wake but EMG shows near-complete skeletal muscle atonia, and sawtooth waves may appear.

Five channels are used, each from a distinct physiological source:

- **EEG Fpz-Cz:** Frontal-central electroencephalogram; primary signal for staging N1–N3 and REM.
- **EEG Pz-Oz:** Parietal-occipital EEG; captures occipital alpha rhythm (8–13 Hz) during relaxed wakefulness.
- **EOG (horizontal):** Electrooculogram recording horizontal eye movements; distinguishes REM eye movements from wakefulness saccades.
- **EMG (submental chin):** Chin electromyogram measuring skeletal muscle tone; key discriminator for REM atonia.
- **Resp (oro-nasal):** Oro-nasal airflow thermistor recording respiratory rate and pattern; aids detection of apnoeic events.

All five channels are recorded at 100 Hz simultaneously but measure different physiological phenomena, making this a **Type 2 (heterogeneous and aligned)** dataset.

After removing transition epochs at recording boundaries, the dataset contains **10,918 30-second epochs** from 78 subjects (153 overnight recordings). Each sample has shape $\mathbb{R}^{5 \times 3000}$ ($C = 5$, $T = 3,000$).

All channels are natively sampled at 100 Hz; no resampling is required. Each 30-second epoch yields $T = 100 \times 30 = 3,000$ time steps. Z-score normalization is applied to each channel independently. Data splits are stratified by subject.

- **PPG-DaLiA.** PPG-DaLiA (PPG Dataset for Life Activities)[30] is a multimodal wearable dataset designed to advance activity recognition and heart-rate estimation under real-world, free-living conditions. It provides synchronized data from a chest-worn device and a wrist-worn device across naturalistic daily activities. Reliable activity recognition from low-cost wrist PPG devices is essential for consumer health wearables and for correcting motion artifacts in continuous heart-rate monitoring.

The task is **nine-class activity classification**. Each sample window is assigned one activity label via majority vote over the 4 Hz label signal within the window:

- **Transient:** Activity-transition segments that do not belong to a stable activity state; the most frequent class (27.8% of training windows).
- **Sitting:** Stationary sedentary posture.
- **Ascending stairs:** Upward stair climbing.
- **Descending stairs:** Downward stair descent.
- **Table soccer:** Upper-limb dominated gaming activity with minimal lower-body motion.
- **Cycling:** Stationary or outdoor pedalling.
- **Driving:** Seated vehicle operation.
- **Lunch break:** Low-activity resting or eating period.
- **Walking:** Moderate-intensity locomotion at a self-selected pace.

Nine channels are drawn from two wearable devices. The wrist device (Empatica E4) contributes BVP/PPG (1 channel), a 3-axis accelerometer (3 channels), electrodermal activity (EDA, 1 channel), and skin temperature (TEMP, 1 channel), totalling 6 channels. The chest device (RespiBAN) contributes a 3-axis accelerometer (3 channels). Native sampling rates differ across sensors (e.g.,

64 Hz for wrist BVP vs. higher rates for chest ACC), and the two devices measure physiologically distinct signals. After resampling to a unified rate, this dataset is **Type 2 (heterogeneous and aligned)**.

The dataset contains 15 subjects. After sliding-window segmentation the dataset yields **64,726 samples**. Each sample has shape $\mathbb{R}^{9 \times 256}$ ($C = 9, T = 256$).

All nine channels are resampled to a unified 32 Hz using downsampling and linear interpolation. An 8-second sliding window with a 2-second stride produces $T = 32 \times 8 = 256$ time steps per sample. Z-score normalization is applied to each channel independently. Data splits are stratified by subject.

- **WESAD.** WESAD (Wearable Stress and Affect Detection)[31] by Schmidt *et al.* is a benchmark for automated, real-time stress and affect recognition from wrist-worn physiological sensors. Stress detection is clinically relevant for mental health monitoring, early burnout prevention, and cardiovascular risk management, especially when deployed passively on consumer wearables without user intervention.

The task is **three-class affective-state classification**:

- **Baseline (Neutral):** Resting state; subjects sit quietly or read neutral materials. This condition establishes each subject’s individual physiological baseline.
- **Stress:** Acute psychological stress induced by the Trier Social Stress Test (TSST), which combines public speaking and mental arithmetic performed in front of an evaluator panel. TSST reliably elevates cortisol and activates the sympathetic nervous system.
- **Amusement:** Mild positive affect induced by a curated selection of short comedy video clips, intended as a low-arousal positive affect condition contrasting with the high-arousal stress condition.

Two wrist physiological channels are used. EDA (Electrodermal Activity) measures skin conductance and reflects sympathetic nervous system arousal; conductance rises within seconds of an acute stressor, making it the most discriminative channel for stress detection. BVP (Blood Volume Pulse / PPG) captures cardiac activity from the wrist; stress-induced sympathetic activation increases heart rate and alters heart-rate variability features embedded in the BVP waveform. Both channels are recorded simultaneously by the Empatica E4 wrist device but reflect different physiological pathways, making this a **Type 2 (heterogeneous and aligned)** dataset.

The dataset contains 15 subjects. After sliding-window segmentation the dataset yields **4,387 samples**. Each sample has shape $\mathbb{R}^{2 \times 480}$ ($C = 2, T = 480$).

We use the wrist EDA and BVP channels only. All channels are resampled to 32 Hz. A 15-second sliding window with a 7.5-second stride produces $T = 32 \times 15 = 480$ time steps. Z-score normalization is applied at the sample level to each channel independently. Data splits are stratified by subject.

Type 3: Heterogeneous and Unaligned Datasets. In this group, modalities differ in both physiological origin and temporal resolution and cannot be placed on a common synchronised time axis without fundamental structural transformation, such as mixing regularly-sampled time series with static embeddings or irregular clinical measurements.

- **MIMIC-IV.** MIMIC-IV (Medical Information Mart for Intensive Care IV)[20] is a comprehensive de-identified critical-care database developed by Beth Israel Deaconess Medical Center (BIDMC) in partnership with MIT. It records the full clinical trajectory of more than 40,000 ICU admissions including vital signs, laboratory values, medications, clinical notes, chest radiographs, and ECG waveforms. In-hospital mortality prediction is a fundamental decision-support problem that informs prognosis, triage, and palliative-care planning.

The task is **binary in-hospital mortality prediction**. The positive label (1) indicates that the patient dies during the current hospital admission; the negative label (0) indicates survival to discharge. The dataset is severely class-imbalanced with an approximate mortality rate of 13%, reflecting real-world ICU mortality distributions.

Four fundamentally heterogeneous modalities are present:

1. **Clinical time series** ($C = 30, T = 48$): Vital signs and laboratory values aggregated into 1-hour bins over the first 48 hours of ICU admission. The 30 channels include heart rate, systolic/diastolic/mean arterial blood pressure, respiratory rate, body temperature, SpO_2 , and key biochemical markers such as glucose, creatinine, potassium, sodium, and bicarbonate.

2. **Chest X-ray features (1024-D static vector):** Visual embeddings pre-extracted from the most recent chest radiograph sourced from MIMIC-CXR-JPG [19], following the multimodal configuration of Han et al. [13], using a pretrained thoracic image encoder, capturing structural lung and cardiac pathology including effusions, cardiomegaly, and consolidations.
3. **ECG features (256-D static vector):** Temporal embeddings pre-extracted from the 12-lead ECG recording closest to ICU admission time, encoding arrhythmia and ischaemia patterns in a compact representation.
4. **Clinical text features (768-D static vector):** Semantic embeddings pre-extracted from clinical notes (nursing notes, discharge summaries) using a pretrained BERT-based clinical language model, encoding free-text observations not captured by structured variables.

Modalities 2–4 are static vectors with no time axis and cannot be aligned with the hourly time series, placing this dataset in **Type 3 (heterogeneous and unaligned)**.

After filtering and preprocessing the dataset contains **5,100 ICU admission episodes**. The time-series component has shape $\mathbb{R}^{30 \times 48}$; the three embedding modalities are fused separately by each model architecture.

Clinical measurements are aggregated into 1-hour bins; missing values within a stay are forward-filled, and any remaining gaps are set to zero. Z-score normalization is applied to the time-series channels using training-split statistics. The embedding vectors are used directly without further normalisation.

- **CinC Challenge 2012 (Challenge-2012).** The PhysioNet [12]/Computing in Cardiology Challenge 2012[35] is a canonical benchmark for ICU mortality prediction, providing irregularly sampled multivariate clinical time series from general medical, cardiac, and surgical ICUs. Its extreme feature sparsity makes it directly relevant to the missingness study in this paper.

The task is **binary ICU mortality prediction**. The positive label (1) indicates in-hospital death; the negative label (0) indicates survival to discharge. The positive (mortality) rate is approximately 13.9%.

Forty-two clinical variables are recorded per patient ($C = 42$), grouped into three categories. General descriptors (quasi-static) include age, gender, height, weight, ICU type (cardiac surgery, medical-surgical, or other), and SAPS-I score. Time-varying vital signs include heart rate, systolic/diastolic/mean arterial blood pressure, respiratory rate, body temperature, $\text{SpO}_2/\text{SaO}_2$, Glasgow Coma Scale (GCS), mechanical ventilation flag, and fraction of inspired oxygen (FiO_2). Time-varying laboratory values include blood urea nitrogen (BUN), creatinine, glucose, potassium (K), sodium (Na), pH, PaO_2 , PaCO_2 , bicarbonate (HCO_3), lactate, total bilirubin, hematocrit (HCT), white blood cell count (WBC), and magnesium (Mg). Clinical measurements are taken at irregular intervals dictated by clinical decisions rather than a fixed schedule. After binning into hourly slots, the resulting 42×48 grid is naturally sparse, with most patient-hours containing no measurements for most variables. This irregular temporal structure combined with the mixture of static and dynamic variables makes this a **Type 3 (heterogeneous and unaligned)** dataset.

The dataset contains approximately **4,000 ICU patient stays** (set-a). After preprocessing each sample has shape $\mathbb{R}^{42 \times 48}$ ($C = 42$, $T = 48$).

Irregular clinical measurements are grouped into 1-hour bins; if a bin contains multiple observations for the same variable, they are averaged. Bins with no observation are set to zero without forward-filling or imputation. This deliberate choice ensures that zero values represent genuine structural missingness rather than imputed estimates, isolating the effect of missing data in a controlled way. Z-score normalization is applied to each of the 42 channels using training-split statistics.

- **CirCor DigiScope (CirCor).** CirCor DigiScope[28] is a cardiac auscultation dataset from the PhysioNet/Computing in Cardiology Challenge 2022, targeting murmur detection from phonocardiogram (PCG, i.e., heart sound) recordings. It covers a large pediatric cardiac screening campaign conducted in Brazil. Early detection of congenital and acquired cardiac abnormalities through heart auscultation is a critical but resource-limited clinical task in low- and middle-income settings, motivating automated screening tools.

The task is **three-class murmur classification**. Label 0 (Absent) indicates that no murmur is detected at any auscultation location and accounts for 75.6% of recordings. Label 1 (Unknown) indicates insufficient audio quality to determine murmur presence (5.0%). Label 2 (Present) indicates a clearly audible murmur at one or more locations (19.4%). The class distribution is severely imbalanced, requiring class-balanced loss weighting.

Each recording is a single-channel PCG waveform captured at one of four standard auscultation sites: Aortic Valve (AV), Pulmonary Valve (PV), Tricuspid Valve (TV), or Mitral Valve (MV). In the preprocessed format, each recording is transformed into a log-mel spectrogram with 64 mel-frequency bins, yielding a time-frequency feature tensor. An additional 14 static demographic and metadata features are broadcast as constant channels across the time dimension and appended to the spectrogram, comprising age group one-hot encoding (6 channels for Neonate, Infant, Child, Adolescent, Young Adult, Unknown), sex (1 channel), height and weight (2 channels), pregnancy status (1 channel), and recording location one-hot encoding (4 channels for AV, PV, TV, MV). The resulting input has $C = 64 + 14 = 78$ channels. The 14 demographic channels are static scalars with no temporal structure; they possess a fundamentally different format and resolution from the time-varying mel spectrogram channels, making temporal alignment between the two channel groups meaningless. This mixture of a dynamic time-frequency modality and static patient-level metadata makes this a **Type 3 (heterogeneous and unaligned)** dataset.

The original collection contains 942 patients and 3,163 WAV recordings at 4,000 Hz. After excluding 45 recordings that exceed the 20-second duration cap, **3,118 recordings** are retained. Each sample has shape $\mathbb{R}^{78 \times 625}$ ($C = 78$, $T = 625$).

Raw PCG waveforms are recorded at 4,000 Hz. Each recording is converted to a log-mel spectrogram using $n_{\text{fft}} = 512$ and $\text{hop_length} = 128$, yielding $T = 625$ time frames for a 20-second clip. Recordings shorter than 20 seconds are zero-padded; longer recordings are truncated. The spectrogram is normalized using global per-channel mean and standard deviation computed from the training split. Static demographic features are broadcast across the T dimension at runtime. Data splits are patient-level (seed=42) to prevent data leakage across recordings from the same patient.

D Protocol Adaptation and Implementation

D.1 Compute Resources

All main experiments are conducted on NVIDIA B200 GPUs (178.4 GiB VRAM) on the University of Florida HiPerGator cluster. Each job occupies one B200 GPU, with approximately three jobs running concurrently. The full benchmark of 810 experimental runs (6 models $\times$ 9 datasets $\times$ 3 seeds $\times$ 5 missing-data conditions) completes in approximately one month of wall-clock time. A small number of early exploratory runs use NVIDIA L4 GPUs (22 GiB); these are superseded by the B200 runs and do not contribute to any reported result.

To ensure a rigorous and fair comparison, we implement a unified evaluation protocol. The primary goal is to ensure that the model architecture is the only variable across all experiments.

D.2 Unified Data Interface

We use a standardized pipeline to ensure all models process the exact same data splits and missingness patterns. We generate unified index files (e.g., `split_train.json`) and a `meta.json` for each dataset. Models load raw data on the fly using these shared indices rather than a single forced intermediate format. After preprocessing, all datasets are saved as shared cache files on disk. All six baseline models read from these exact same cached files, guaranteeing absolute data-level fairness across the entire evaluation.

D.3 Framework-Agnostic Missingness Library

We implement missingness injection as a single Python module that is copied into all six model projects. This ensures that the mathematical logic for modality dropping and time-block masking is byte-for-byte identical across all frameworks, regardless of each project’s underlying training infrastructure.

Each sample is represented as a dict with four fields:

- $\mathbf{x} \in \mathbb{R}^{C \times T}$: raw channel signals.
- $\text{mask} \in \{0, 1\}^{C \times T}$: per-timestep validity mask (1 = valid, 0 = padded or missing).
- $\text{mod_mask} \in \{0, 1\}^C$: per-channel presence mask (1 = present, 0 = dropped).

- $\text{ts_mod} \in \{\text{True}, \text{False}\}^C$: flags marking each channel as a time-series modality. Channels with $\text{ts_mod} = \text{False}$ are skipped by the time-block injector.

The injectors update `mask` and `mod_mask` in place. Downstream model code treats these fields either by multiplying the input tensor by the mask before projection or by passing it as an attention mask.

D.3.1 The `__getitem__` Interface

The unified entry point is `apply_missingness(sample, mode, *, p_mod, block_n, block_m, block_n_max, seed, sample_id, min_kept)`, which dispatches on `mode` $\in \{\text{none}, \text{modality}, \text{block}\}$. It is called inside `Dataset.__getitem__` immediately after the raw sample is loaded from the cache, so missingness is injected on-the-fly during data loading rather than written to disk. This allows any missing rate to be evaluated without storing additional copies of the data. All six model projects follow the same call pattern, so models with entirely different internal formats (channel-independent projections, cross-modal attention, 1-D CNN encoders) receive the identical missingness pattern without any format-specific modification to the library.

D.3.2 Deterministic Stochasticity

For a given pair $(\text{seed}, \text{sample_id})$, the generated missingness pattern is identical across all models and runs. Before each injection, a local random generator is constructed. The multiplicative folding ensures that nearby sample indices produce uncorrelated patterns. Because the same formula is used in every model project, the mask applied to sample i under a given configuration is identical whether the sample is loaded by CLIMB, MIRA, or any other model, guaranteeing a fair cross-model comparison.

D.4 Implementation of Missingness Modes

D.4.1 Modality Missing

Our implementation of **modality missing** is a channel-level missing completely at random (MCAR) simulation in which each input channel is independently dropped to approximate whole-sensor failures. We use this channel-level approach because it provides a controlled and reproducible proxy for modality-level loss across all nine datasets, which differ widely in channel count and channel type. We note that this simulation does not always match the strict semantic meaning of a modality. In PTB-XL, for example, each of the 12 ECG leads measures the same cardiac activity from a different electrode position, so dropping one lead is more precisely a channel-level event than the loss of an independent modality. In CirCor, removing a mel-spectrogram frequency bin removes part of a single audio representation rather than an entire sensor source. A stricter protocol would group channels by their physical source and drop all channels from that source at once, capturing correlated source-level failures. We treat this source-level correlated missingness as a known limitation of the current simulation and leave it as future work.

For each currently present channel ($\text{mod_mask}[c] = 1$), the injector performs an independent Bernoulli trial with probability p_{mod} . A successful trial sets $\text{mod_mask}[c] \leftarrow 0$ and $\text{mask}[c, :] \leftarrow \mathbf{0}$, zeroing the entire channel row in the validity mask. A safeguard enforces $\text{min_kept} = 1$: if the Bernoulli draws would drop all present channels, the injector randomly restores one dropped channel so the model always receives at least one valid input. Under this strategy, the realized number of dropped channels follows approximately $\text{Binomial}(n_{\text{present}}, p_{\text{mod}})$.

D.4.2 Within-Modality Missing

This mode carves non-overlapping contiguous segments independently in each time-series channel, simulating asynchronous sensor interruptions. For a channel of length T , the procedure is:

1. Compute the block length range: $\ell_{\min} = \lceil \text{block_n} \cdot T \rceil$, $\ell_{\max} = \lceil \text{block_n_max} \cdot T \rceil$. We use $\text{block_n} = 0.05$ and $\text{block_n_max} = 0.10$, so each block covers 5–10% of T .
2. Estimate the number of blocks required to cover fraction block_m of the sequence:

$$k = \left\lceil \frac{\text{block_m} \cdot T}{(\ell_{\min} + \ell_{\max})/2} \right\rceil.$$

3. For each block, uniformly sample a start position and check for overlap with already-placed blocks. If an overlap is found, resample up to 64 times; if no valid position is found, stop placing further blocks for this channel.
4. Set $\text{mask}[c, \text{start} : \text{end}] \leftarrow 0$ for each placed block.

Each channel uses an independent sub-generator: before iterating over channels, the shared `rng` draws one 64-bit seed per channel upfront, and each channel’s block placement proceeds from its own `np.random.default_rng`. This ensures that different channels miss different time windows while the entire per-sample pattern remains reproducible from $(\text{seed}, \text{sample_id})$.

D.5 Standardized Training Protocols

Loss functions and class imbalance. For multi-class tasks (Sleep-EDF, PPG-DaLiA, WESAD, CirCor), we use cross-entropy with inverse-frequency class weights. For binary tasks (HAR-UP, MIMIC-IV, Challenge-2012), we use binary cross-entropy or two-class cross-entropy (depending on the model architecture) with a scalar positive-class weight. For multi-label tasks (PTB-XL, Chapman-Shaoxing), we use `BCEWithLogitsLoss` with a per-class positive weight:

$$w_{\text{pos}} = n_{\text{neg}} / n_{\text{pos}},$$

computed from the training split. This is particularly important for clinical datasets with severe class imbalance, such as MIMIC-IV (positive rate $\approx 13\%$, $w_{\text{pos}} \approx 6.6$) and Challenge-2012 (positive rate $\approx 13.9\%$, $w_{\text{pos}} \approx 6.2$).

Checkpoint selection. All six models select the best checkpoint by the highest validation Macro-AUROC score, regardless of the primary test metric. This common stopping criterion avoids per-model tuning of the selection rule and ensures that reported test results reflect each model at its peak validation state.

Maestro training protocol. Maestro diverges from the standard protocol only during training. Rather than receiving externally injected missing masks, Maestro applies its own curriculum modality dropout internally: the per-channel drop probability starts at 0 and increases linearly to a maximum of 40% over the first 10 warmup epochs, after which it remains fixed. This schedule encourages the model to gradually adapt to missing inputs without being exposed to severe dropout from the start. Critically, this internal dropout is applied only during the forward pass of training batches and is disabled at inference time. For validation and test evaluation, Maestro receives the same externally injected missing masks as all other models, generated from the shared $(\text{seed}, \text{sample_id})$ deterministic protocol. This means that all reported validation and test AUROC numbers for Maestro are directly comparable to those of other models under identical missing conditions.

E Detailed Result

E.1 Per-Dataset Per-Run Results

This section reports full numerical results for all nine datasets across all five missing-data settings and six models. Every experiment is repeated under three independent random seeds: `r1` (seed=42), `r2` (seed=2026), and `r3` (seed=114514), and each run is listed separately so that run-to-run variance is visible. For each seed and setting, we report two metrics: AUROC (macro one-vs-rest) and Macro-F1. The five settings are Clean (no missing data), Modality 20% and 50% (channel-level MCAR at the respective drop probability), and Within-modality 20% and 50% (block-level masking covering 20% or 50% of each channel’s time steps). Bold entries mark the best AUROC and best Macro-F1 within each setting row. The superscript $\dagger$ marks ShaSpec runs on Challenge-2012 where training collapsed to predicting a single class ($F1=0$); $\ddagger$ marks isolated CLIMB or MIRA runs that similarly collapsed to all-positive prediction. Full results for each of the nine datasets are presented in Tables 10–18.

Table 10: HAR-UP: AUC and Macro-F1 (AUC: AUROC) (binary fall detection).

Setting	Run	CLIMB		MIRA		Flex-MoE		ShaSpec		FuseMoE		Maestro	
		AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1
Clean	r1	.989	.934	.983	.911	.856	.766	.889	.871	.914	.750	.995	.934
	r2	.990	.923	.970	.830	.844	.771	.924	.892	.947	.882	.996	.962
	r3	.987	.900	.976	.892	.865	.804	.930	.892	.951	.844	.981	.935
Mod 20%	r1	.988	.946	.980	.907	.778	.710	.888	.876	.883	.796	.984	.941
	r2	.985	.941	.943	.859	.815	.702	.917	.892	.884	.791	.989	.940
	r3	.981	.909	.975	.898	.841	.749	.915	.892	.880	.780	.989	.924
Mod 50%	r1	.977	.878	.892	.796	.682	.601	.878	.855	.768	.668	.956	.887
	r2	.978	.887	.966	.898	.764	.714	.896	.876	.781	.701	.964	.844
	r3	.979	.914	.950	.870	.811	.713	.888	.865	.790	.721	.965	.903
Block 20%	r1	.991	.934	.983	.935	.835	.742	.874	.849	.859	.797	.990	.940
	r2	.975	.903	.965	.892	.815	.766	.887	.855	.870	.754	.997	.946
	r3	.984	.919	.960	.876	.829	.718	.887	.849	.868	.795	.987	.935
Block 50%	r1	.916	.876	.959	.854	.775	.673	.857	.833	.863	.790	.956	.747
	r2	.962	.868	.943	.839	.840	.763	.867	.817	.832	.761	.977	.776
	r3	.941	.786	.910	.796	.769	.695	.874	.810	.830	.718	.979	.882

Table 11: PTB-XL: AUC and Macro-F1 (AUC: AUROC) (multi-label, 5 classes).

Setting	Run	CLIMB		MIRA		Flex-MoE		ShaSpec		FuseMoE		Maestro	
		AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1
Clean	r1	.853	.546	.831	.504	.771	.456	.860	.630	.663	.191	.884	.613
	r2	.832	.538	.820	.460	.763	.424	.867	.637	.709	.258	.881	.631
	r3	.829	.553	.839	.523	.782	.518	.868	.638	.660	.180	.883	.619
Mod 20%	r1	.832	.502	.825	.512	.719	.418	.856	.626	.658	.333	.873	.595
	r2	.821	.507	.828	.497	.737	.475	.862	.632	.666	.295	.870	.614
	r3	.814	.525	.822	.487	.757	.462	.861	.632	.624	.181	.876	.625
Mod 50%	r1	.811	.523	.828	.504	.714	.464	.843	.611	.638	.251	.845	.536
	r2	.800	.480	.792	.445	.703	.464	.845	.613	.574	.225	.848	.569
	r3	.799	.485	.818	.506	.703	.455	.850	.616	.595	.222	.852	.596
Block 20%	r1	.824	.503	.847	.535	.740	.444	.858	.626	.694	.262	.877	.615
	r2	.823	.531	.831	.527	.723	.400	.865	.638	.672	.223	.875	.629
	r3	.819	.524	.829	.501	.745	.442	.865	.637	.622	.224	.879	.617
Block 50%	r1	.795	.446	.811	.459	.714	.419	.854	.625	.614	.135	.852	.522
	r2	.797	.452	.831	.525	.701	.389	.861	.634	.600	.122	.855	.523
	r3	.803	.474	.812	.450	.703	.404	.860	.632	.609	.217	.860	.497

Table 12: Chapman-Shaoxing: AUC and Macro-F1 (AUC: AUROC) (multi-label, 7 classes).

Setting	Run	CLIMB		MIRA		Flex-MoE		ShaSpec		FuseMoE		Maestro	
		AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1
Clean	r1	.793	.332	.839	.396	.729	.379	.845	.496	.678	.322	.772	.433
	r2	.775	.323	.818	.360	.725	.354	.850	.503	.660	.356	.774	.421
	r3	.790	.334	.824	.409	.728	.355	.854	.507	.668	.315	.867	.432
Mod 20%	r1	.789	.324	.817	.391	.693	.367	.845	.498	.642	.324	.761	.427
	r2	.776	.322	.837	.439	.705	.347	.850	.506	.646	.330	.755	.398
	r3	.784	.325	.839	.412	.735	.329	.851	.502	.626	.358	.856	.420
Mod 50%	r1	.776	.320	.808	.394	.686	.328	.839	.494	.579	.307	.726	.398
	r2	.764	.306	.812	.418	.682	.348	.841	.496	.595	.291	.717	.393
	r3	.777	.316	.811	.379	.675	.358	.850	.526	.565	.276	.832	.382
Block 20%	r1	.779	.311	.798	.390	.715	.346	.839	.488	.636	.288	.753	.375
	r2	.769	.307	.832	.415	.700	.340	.840	.493	.623	.333	.770	.327
	r3	.783	.321	.810	.395	.702	.342	.848	.499	.644	.349	.852	.405
Block 50%	r1	.742	.240	.804	.344	.663	.320	.828	.478	.584	.276	.732	.238
	r2	.724	.230	.807	.365	.676	.318	.837	.495	.603	.318	.732	.338
	r3	.758	.268	.816	.404	.666	.327	.841	.487	.611	.301	.792	.310

Table 13: Sleep-EDF: AUC and Macro-F1 (AUC: AUROC) (5-class, multiclass).

Setting	Run	CLIMB		MIRA		Flex-MoE		ShaSpec		FuseMoE		Maestro	
		AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1
Clean	r1	.983	.754	.958	.629	.971	.692	.984	.762	.929	.577	.969	.695
	r2	.976	.687	.957	.629	.975	.715	.984	.765	.930	.586	.972	.702
	r3	.983	.744	.932	.567	.974	.697	.984	.761	.935	.596	.971	.697
Mod 20%	r1	.973	.691	.930	.563	.963	.660	.980	.738	.891	.506	.955	.658
	r2	.972	.691	.920	.558	.960	.646	.980	.735	.889	.497	.950	.634
	r3	.973	.682	.928	.560	.962	.656	.979	.737	.890	.513	.951	.643
Mod 50%	r1	.946	.606	.885	.466	.924	.575	.958	.662	.791	.371	.889	.521
	r2	.949	.618	.875	.466	.925	.565	.961	.659	.803	.390	.886	.528
	r3	.945	.597	.852	.437	.920	.560	.961	.669	.797	.389	.893	.520
Block 20%	r1	.981	.733	.945	.594	.964	.661	.982	.740	.906	.545	.955	.629
	r2	.980	.737	.926	.569	.969	.674	.983	.748	.917	.558	.955	.624
	r3	.978	.728	.956	.618	.969	.691	.984	.753	.920	.563	.958	.622
Block 50%	r1	.974	.705	.917	.552	.962	.642	.981	.741	.891	.511	.897	.446
	r2	.969	.695	.927	.570	.963	.674	.981	.740	.879	.492	.894	.450
	r3	.975	.694	.937	.583	.957	.636	.982	.738	.891	.511	.889	.433

Table 14: PPG-DaLiA: AUC and Macro-F1 (AUC: AUROC) (9-class HAR).

Setting	Run	CLIMB		MIRA		Flex-MoE		ShaSpec		FuseMoE		Maestro	
		AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1
Clean	r1	.945	.649	.937	.672	.929	.628	.959	.731	.924	.604	.956	.765
	r2	.939	.670	.941	.681	.928	.620	.959	.740	.927	.641	.960	.773
	r3	.941	.683	.932	.653	.932	.646	.959	.735	.930	.636	.953	.748
Mod 20%	r1	.934	.668	.935	.660	.909	.576	.953	.714	.905	.566	.928	.662
	r2	.931	.633	.929	.665	.909	.563	.953	.707	.897	.557	.934	.688
	r3	.924	.636	.936	.663	.907	.573	.953	.713	.900	.564	.933	.674
Mod 50%	r1	.911	.568	.916	.585	.880	.484	.935	.648	.874	.471	.857	.497
	r2	.903	.551	.907	.589	.871	.469	.936	.644	.872	.454	.875	.507
	r3	.899	.541	.913	.592	.878	.484	.936	.644	.863	.439	.861	.500
Block 20%	r1	.939	.621	.934	.670	.913	.586	.947	.670	.926	.615	.942	.708
	r2	.931	.643	.937	.673	.920	.595	.952	.694	.910	.588	.944	.675
	r3	.932	.658	.938	.672	.922	.617	.952	.699	.923	.610	.944	.720
Block 50%	r1	.897	.563	.936	.659	.908	.561	.936	.602	.919	.577	.896	.518
	r2	.911	.583	.932	.635	.907	.555	.940	.628	.914	.569	.885	.478
	r3	.913	.572	.933	.658	.901	.550	.941	.630	.909	.556	.865	.406

Table 15: WESAD: AUC and Macro-F1 (AUC: AUROC) (3-class affect recognition).

Setting	Run	CLIMB		MIRA		Flex-MoE		ShaSpec		FuseMoE		Maestro	
		AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1
Clean	r1	.704	.505	.655	.448	.691	.437	.671	.454	.572	.364	.719	.449
	r2	.697	.493	.720	.509	.683	.415	.703	.479	.584	.357	.751	.526
	r3	.703	.505	.678	.453	.676	.451	.649	.419	.593	.386	.739	.506
Mod 20%	r1	.650	.443	.636	.425	.691	.424	.641	.434	.563	.360	.711	.473
	r2	.675	.459	.611	.406	.681	.431	.641	.446	.572	.356	.726	.489
	r3	.672	.461	.665	.456	.673	.437	.643	.423	.567	.365	.726	.461
Mod 50%	r1	.626	.430	.590	.415	.668	.399	.616	.428	.584	.377	.680	.472
	r2	.661	.439	.644	.439	.678	.410	.673	.490	.604	.387	.695	.456
	r3	.662	.463	.628	.427	.678	.435	.666	.426	.587	.356	.713	.484
Block 20%	r1	.690	.499	.652	.436	.697	.432	.646	.409	.601	.389	.716	.439
	r2	.696	.455	.670	.468	.709	.437	.655	.436	.565	.367	.746	.481
	r3	.703	.496	.669	.459	.692	.448	.677	.443	.601	.366	.717	.461
Block 50%	r1	.664	.434	.629	.412	.671	.393	.627	.406	.582	.374	.673	.406
	r2	.653	.462	.609	.399	.690	.420	.636	.426	.578	.350	.701	.407
	r3	.652	.447	.626	.421	.681	.421	.655	.414	.598	.361	.675	.425

Table 16: MIMIC-IV: AUC and Macro-F1 (AUC: AUROC) (binary IHM-48, 4 modalities).

Setting	Run	CLIMB		MIRA		Flex-MoE		ShaSpec		FuseMoE		Maestro	
		AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1
Clean	r1	.789	.392	.760	.392	.754	.389	.657	.313	.813	.395	.813	.355
	r2	.781	.430	.711	.360	.761	.408	.658	.318	.809	.425	.795	.410
	r3	.776	.413	.753	.361	.766	.413	.655	.303	.813	.441	.808	.414
Mod 20%	r1	.755	.353	.726	.368	.737	.377	.638	.309	.779	.397	.781	.393
	r2	.747	.357	.720	.345	.733	.369	.633	.292	.771	.348	.773	.392
	r3	.749	.373	.729	.380	.742	.384	.642	.295	.782	.410	.777	.401
Mod 50%	r1	.710	.360	.685	.340	.694	.343	.615	.279	.732	.349	.733	.358
	r2	.703	.310	.675	.334	.688	.333	.607	.290	.736	.345	.727	.359
	r3	.712	.351	.681	.339	.705	.348	.623	.248	.741	.345	.726	.362
Block 20%	r1	.776	.403	.756	.389	.756	.398	.653	.304	.803	.438	.804	.424
	r2	.774	.413	.761	.403	.759	.392	.654	.312	.806	.438	.785	.387
	r3	.773	.396	.753	.379	.762	.404	.653	.310	.805	.375	.806	.409
Block 50%	r1	.753	.363	.730	.375	.740	.356	.645	.305	.794	.371	.796	.394
	r2	.755	.397	.750	.359	.752	.391	.645	.315	.791	.380	.793	.432
	r3	.747	.378	.751	.338	.758	.403	.645	.307	.796	.373	.798	.409

Table 17: Challenge-2012: AUC and Macro-F1 (AUC: AUROC) (binary ICU mortality). †: ShaSpec collapse (all-negative / all-positive prediction; F1≈0). ‡: CLIMB or MIRA collapses to all-positive in that run.

Setting	Run	CLIMB		MIRA		Flex-MoE		ShaSpec		FuseMoE		Maestro	
		AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1
Clean	r1	.691	.297	.753	.396	.767	.411	.662 [†]	.000 [†]	.809	.427	.801	.425
	r2	.709	.331	.747	.375	.758	.367	.545 [†]	.244 [†]	.802	.421	.815	.437
	r3	.692	.339	.726	.378	.778	.402	.624 [†]	.000 [†]	.807	.422	.819	.421
Mod 20%	r1	.661	.311	.734	.306	.751	.392	.686 [†]	.000 [†]	.678	.322	.772	.379
	r2	.636 [‡]	.244 [‡]	.710	.338	.741	.367	.611 [†]	.000 [†]	.694	.349	.763	.394
	r3	.623 [‡]	.244 [‡]	.727	.321	.764	.389	.638 [†]	.000 [†]	.705	.323	.797	.403
Mod 50%	r1	.666	.307	.692	.324	.734	.357	.636 [†]	.000 [†]	.637	.256	.706	.307
	r2	.667	.330	.676 [‡]	.247 [‡]	.690	.319	.607 [†]	.245 [†]	.610	.263	.706	.270
	r3	.667	.306	.723	.322	.728	.367	.579 [†]	.258 [†]	.601	.251	.755	.358
Block 20%	r1	.696	.344	.714	.344	.755	.382	.598 [†]	.000 [†]	.810	.398	.803	.436
	r2	.663 [‡]	.244 [‡]	.730	.323	.748	.373	.644 [†]	.000 [†]	.786	.376	.815	.440
	r3	.681	.309	.742	.356	.769	.406	.672 [†]	.018 [†]	.803	.404	.823	.423
Block 50%	r1	.672	.330	.728	.357	.739	.328	.656 [†]	.276 [†]	.795	.414	.786	.406
	r2	.668	.290	.695	.330	.734	.354	.608 [†]	.000 [†]	.784	.416	.785	.395
	r3	.683	.317	.722	.385	.754	.383	.620 [†]	.000 [†]	.799	.409	.798	.379

Table 18: CirCor: AUC and Macro-F1 (AUC: AUROC) (3-class murmur detection).

Setting	Run	CLIMB		MIRA		Flex-MoE		ShaSpec		FuseMoE		Maestro	
		AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1
Clean	r1	.670	.471	.610	.381	.574	.365	.672	.472	.667	.382	.802	.591
	r2	.639	.442	.599	.394	.556	.354	.698	.424	.679	.405	.791	.566
	r3	.638	.429	.595	.388	.571	.351	.664	.437	.619	.438	.794	.563
Mod 20%	r1	.663	.452	.618	.433	.504	.305	.658	.445	.565	.376	.787	.570
	r2	.647	.433	.618	.408	.521	.292	.676	.443	.599	.399	.794	.558
	r3	.645	.400	.551	.343	.563	.349	.644	.441	.633	.387	.767	.554
Mod 50%	r1	.650	.447	.574	.397	.543	.314	.683	.443	.609	.407	.758	.500
	r2	.635	.414	.547	.375	.499	.335	.700	.446	.570	.402	.769	.525
	r3	.649	.437	.559	.361	.576	.364	.648	.425	.638	.418	.737	.497
Block 20%	r1	.672	.442	.635	.443	.568	.331	.647	.406	.601	.383	.757	.545
	r2	.650	.434	.608	.420	.579	.351	.659	.402	.677	.388	.783	.554
	r3	.655	.450	.632	.447	.567	.350	.631	.358	.614	.427	.783	.545
Block 50%	r1	.642	.408	.543	.357	.568	.340	.680	.389	.594	.382	.766	.471
	r2	.618	.444	.564	.364	.596	.378	.665	.379	.673	.369	.759	.500
	r3	.637	.436	.589	.379	.554	.339	.644	.337	.602	.439	.752	.461

E.2 Performance Degradation by Model Family

Tables 19 and 20 provide a quantitative breakdown of performance degradation grouped by architecture family across all nine datasets and both missingness modes. We assign the six models to three families based on their fusion design: channel-independent (Chan.-Indep.: CLIMB and MIRA), shared-specific (Shared-Spec.: ShaSpec), and mixture-of-experts fusion (MoE-Fusion: Flex-MoE, FuseMoE, and Maestro). For each model m , dataset d , and missing rate $r \in \{20\%, 50\%\}$, we first compute per-model degradation $\delta_{m,d,r} = \text{AUROC}_{\text{clean}} - \text{AUROC}_{\text{missing}}$, where each term is the mean test AUROC over three independent seeds ($s \in \{42, 2026, 114514\}$). The value reported in each cell is then the mean of $\delta_{m,d,r}$ over all models belonging to that family. A positive entry indicates that the family loses that many AUROC points relative to the clean baseline; a negative entry indicates a marginal improvement under missing conditions, which can occur when the clean baseline is itself unstable due to class imbalance or small dataset size, as seen for Shared-Spec. on Challenge-2012. Across both tables, MoE-Fusion models tend to show larger degradation than channel-independent or shared-specific models in most settings, especially at the 50% missing rate, which quantitatively supports the claim that architecture family is the strongest predictor of robustness to missing data.

Table 19: Mean ΔAUROC (clean – missing) by model family under modality missing. Chan.-Indep.: CLIMB and MIRA; Shared-Spec.: ShaSpec; MoE-Fusion: Flex-MoE, FuseMoE, and Maestro. Larger values indicate greater performance drop.

Dataset	Rate	Chan.-Indep.	Shared-Spec.	MoE-Fusion
Sleep-EDF	20%	.015	.005	.024
	50%	.056	.024	.089
PTB-XL	20%	.011	.005	.024
	50%	.026	.019	.058
Chapman	20%	.000	.001	.020
	50%	.015	.006	.061
MIMIC-IV	20%	.024	.019	.029
	50%	.067	.041	.072
Challenge	20%	.038	-.035	.053
	50%	.038	.003	.113
PPG-DaLiA	20%	.008	.006	.024
	50%	.031	.023	.067
WESAD	20%	.041	.033	.011
	50%	.058	.023	.014
HAR-UP	20%	.007	.007	.034
	50%	.026	.027	.096
CirCor	20%	.001	.019	.036
	50%	.023	.001	.040

F Detailed Diffusion Imputation Results

F.1 Imputation Model and Pipeline

We train a conditional score-based diffusion model [14, 39] on PTB-XL independently of all benchmark classifiers. The model operates at native ECG resolution ($T = 1000, 100 \text{ Hz} \times 10 \text{ s}$) across all 12 leads and is conditioned on observed channels to fill missing segments on a per-channel basis, with no cross-lead conditioning.

Training. Missing regions are simulated with a contiguous block pattern: each affected channel has one or more blocks zeroed out, where each block spans 5–10% of the signal length (50–100 timesteps), and the total missing fraction per channel is set to 20% or 50%. The model is trained for 20 epochs with batch size 16 and 50 diffusion steps (β linearly spaced from 10^{-4} to 0.02). The architecture uses 4 diffusion layers, 64 channels, 8 attention heads, and embedding dimensions of 128 (time) and 16 (feature).

Offline imputation pipeline. After training, the model imputes the entire PTB-XL train, validation, and test splits and saves the results as pre-computed arrays. Each benchmark classifier loads these

Table 20: Mean Δ AUROC (clean – missing) by model family under within-modality missing. Family definitions follow Table 19.

Dataset	Rate	Chan.-Indep.	Shared-Spec.	MoE-Fusion
Sleep-EDF	20%	.004	.001	.013
	50%	.015	.003	.045
PTB-XL	20%	.005	.002	.019
	50%	.026	.007	.054
Chapman	20%	.011	.007	.022
	50%	.031	.014	.058
MIMIC-IV	20%	-.004	.003	.005
	50%	.014	.011	.013
Challenge	20%	.016	-.028	.005
	50%	.025	-.018	.020
PPG-DaLiA	20%	.004	.008	.010
	50%	.019	.020	.037
WESAD	20%	.013	.015	-.004
	50%	.054	.035	.018
HAR-UP	20%	.006	.032	.033
	50%	.044	.049	.059
CirCor	20%	-.017	.032	.014
	50%	.026	.015	.021

pre-imputed signals directly, with on-the-fly corruption disabled. This two-stage design keeps the imputation and classification training processes fully decoupled.

Modality-missing condition. For the modality-missing experiment, the same within-channel model is applied to leads that are entirely absent. This is an out-of-distribution use: the model is trained to fill short intra-channel blocks and has no observed signal to condition on when a full lead group is dropped. The resulting reconstruction is drawn from an unconstrained prior, which explains the performance degradation reported in Table 7.

F.2 Per-Run Results

This section provides per-run details for the diffusion-based imputation experiment described in Section 4.5. Three downstream classifiers (CLIMB, Flex-MoE, ShaSpec) are evaluated under both modality missing and within-modality missing conditions, across three independent seeds. Table 21 shows per-run results under modality missing; Table 22 shows per-run results under within-modality missing. Notation: **Raw**: model trained on zero-filled corrupted input. **Diff.**: model trained on diffusion-reconstructed input.

Cross-seed stability. Comparing the three seeds reveals a clear pattern of consistency. For CLIMB and Flex-MoE, the imputation improvement over raw missing is present in all nine seed–rate combinations; no seed shows a regression. The single largest gain for Flex-MoE occurs at seed=114514, reaching Δ AUROC $\approx +0.047$ at both rates (20%: $0.7694 - 0.7225$; 50%: $0.7478 - 0.7006$), explaining the relatively high variance in the F1 column for that model at 20% (± 0.026). For ShaSpec, the improvements are uniformly small (Δ AUROC < 0.010 in every individual run), confirming that shared-specific decomposition already exploits cross-lead redundancy effectively, leaving little room for diffusion-based reconstruction to add value. Across all 18 within-modality per-run comparisons (3 models $\times$ 2 rates $\times$ 3 seeds), diffusion-based imputation does not degrade AUROC relative to the matched raw-missing baseline in any of the evaluated runs. Note that this holds for within-modality missing in this PTB-XL experiment; under modality missing, imputation degrades performance across all evaluated settings (Table 7), as discussed in Section 4.5.

Table 21: Per-run modality missing vs. diffusion-based modality imputation on PTB-XL, three seeds. **Raw**: model trained on zero-filled dropped-modality input. **Diff.**: same model trained on diffusion-reconstructed input. Δ denotes performance change from Raw to Diff. **AUC**: AUROC; **F1**: Macro-F1. r1=42, r2=2026, r3=114514.

Model	Run	Rate	Raw		Diff.		Δ	
			AUC	F1	AUC	F1	AUC	F1
CLIMB	r1	20%	.832	.502	.815	.494	-.017	-.009
		50%	.811	.523	.790	.428	-.021	-.095
	r2	20%	.821	.507	.816	.510	-.005	+.003
		50%	.800	.480	.801	.454	+.001	-.026
	r3	20%	.814	.525	.817	.504	+.003	-.021
		50%	.799	.485	.797	.461	-.002	-.024
Flex-MoE	r1	20%	.719	.418	.722	.391	+.003	-.027
		50%	.714	.464	.689	.358	-.026	-.107
	r2	20%	.737	.475	.725	.404	-.011	-.071
		50%	.703	.464	.676	.337	-.027	-.127
	r3	20%	.757	.462	.743	.421	-.014	-.041
		50%	.703	.455	.670	.327	-.033	-.128
ShaSpec	r1	20%	.856	.626	.852	.621	-.004	-.005
		50%	.843	.611	.826	.599	-.018	-.013
	r2	20%	.862	.632	.850	.623	-.012	-.009
		50%	.845	.613	.823	.597	-.022	-.016
	r3	20%	.861	.632	.852	.620	-.009	-.012
		50%	.850	.616	.826	.599	-.024	-.017

Table 22: Per-run within-modality missing vs. diffusion-based imputation on PTB-XL, three seeds. **Raw**: model trained on zero-filled corrupted input. **Diff.**: same model trained on diffusion-reconstructed input. Δ denotes performance change from Raw to Diff. **AUC**: AUROC; **F1**: Macro-F1. r1=42, r2=2026, r3=114514.

Model	Run	Rate	Raw		Diff.		Δ	
			AUC	F1	AUC	F1	AUC	F1
CLIMB	r1	20%	.824	.503	.827	.530	+.002	+.027
		50%	.795	.446	.813	.484	+.019	+.039
	r2	20%	.823	.531	.828	.523	+.005	-.008
		50%	.798	.452	.816	.493	+.019	+.041
	r3	20%	.819	.524	.822	.526	+.004	+.001
		50%	.803	.474	.807	.482	+.005	+.009
Flex-MoE	r1	20%	.740	.444	.758	.413	+.017	-.032
		50%	.714	.419	.743	.420	+.029	+.001
	r2	20%	.745	.442	.752	.429	+.006	-.013
		50%	.703	.404	.741	.420	+.039	+.017
	r3	20%	.723	.400	.769	.463	+.047	+.063
		50%	.701	.389	.748	.442	+.047	+.053
ShaSpec	r1	20%	.858	.626	.867	.636	+.009	+.010
		50%	.854	.625	.858	.630	+.004	+.005
	r2	20%	.865	.638	.866	.636	+.001	-.002
		50%	.861	.634	.860	.632	.000	-.002
	r3	20%	.865	.637	.867	.637	+.002	.000
		50%	.860	.632	.860	.632	.000	.000

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